

Locked in, levered up: Risk, return, and ruin in DeFi lending[☆]

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ABSTRACT

Using comprehensive transaction level loan data for the MakerDAO protocol (2019–2023), this study investigates decentralized finance (DeFi) lending dynamics, focusing on the determinants of loan demand and the interplay between leverage, skill, and user performance. We document a counterintuitive positive relationship between the cost of borrowing and loan demand, consistent with yield seeking behavior. Moreover, blockchain- and protocol-specific frictions, such as gas fees shape borrowing activity. At the vault level, leverage universally reduces returns and amplifies liquidation risk, with unskilled users incurring significantly greater losses than skilled counterparts under extreme leverage. While skilled users mitigate moderate leverage risks through active management, excessive leverage erodes performance across all skill levels, with forced liquidations accounting for a significant proportion of this decline. The findings reveal critical trade-offs in DeFi permissionless architecture. While skilled participants exploit leverage strategically, systemic design features disproportionately penalize less sophisticated users.

1. Introduction

Decentralized Finance (DeFi) is a rapidly evolving financial architecture, leveraging smart contract automation on permissionless blockchains to offer financial services with reduced intermediation. Among its fastest-growing segments are lending protocols, which now account for over \$50 billion in total value locked (TVL). A central feature of these protocols is the ability for users to borrow against crypto-asset collateral, governed by pre-programmed rules for critical parameters like interest rates and liquidation thresholds. Given the novelty of these markets, which are dominated by retail investors, understanding the capacity for users to attain highly leveraged positions and the impact of these positions on users' portfolios are highly relevant for informed discussions of investor protection, market integrity, and the innovative development of decentralized lending systems.

The principles of collateralized borrowing and financial leverage are well-established in traditional finance, where their significance has been extensively analyzed through multiple theoretical frameworks. These include general equilibrium models (Geanakoplos, 2001, 2010), the theory of financial intermediaries as reflected in the behaviors of key market participants (Adrian & Shin, 2010, 2014), and the micro-foundations of leverage decisions under asymmetric information (Acharya & Viswanathan, 2011). Building on these theoretical foundations and leveraging access to transaction-level loan data, our focus in this paper is two-fold –

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exploring the determinants of aggregate loan demand in decentralized financial markets; and examining the role of leverage and skill on financial performance.

Within DeFi, two primary economic models for collateralized borrowing have emerged. The first, typified by protocols such as Aave and Compound, operates as a peer-to-peer intermediation market for loanable funds. In this model, borrowers draw loans from liquidity pools by pledging crypto-assets as collateral, with interest rates often determined algorithmically based on the supply and demand dynamics within each pool. The second model, which is the central focus of this paper and is exemplified by the MakerDAO protocol, enables the endogenous creation of protocol-native debt liabilities through permissionless collateralized debt positions ('vaults').¹ Here, participants do not borrow from pre-existing pools of user-supplied capital. Instead, they deposit approved crypto-assets into a vault to mint protocol-defined debt obligations, typically in the form of a native decentralized stablecoin. The protocol itself functions as the direct facility for this debt issuance, governed by a minimum collateralization ratio determined through its decentralized governance mechanism.

Given the fundamental differences between these collateralized loan models, we utilize loan-level data to examine factors affecting aggregate loan demand in decentralized markets. Prior research has primarily focused on liquidity pool-based lending structures and their associated incentive mechanisms (Cornelli et al., 2024; Saengchote, 2023), which differ significantly from the vault-based model examined in this paper. These liquidity mining programs, however, as discussed by Park & Stinner (2023) incentivize users with protocol tokens (e.g., COMP, AAVE) for both supplying and borrowing the same asset, fostering a reward-farming loop. In contrast, the Maker Protocol does not inherently incorporate such liquidity bootstrapping incentives. This absence of protocol-native token rewards fosters a more stable and organic borrowing environment, enabling a clearer identification of the fundamental drivers of decentralized loan demand.

Our findings reveal several interesting results. Despite the absence of direct governance token rewards that characterize peer-to-peer DeFi protocols, we show a positively significant association between the cost of borrowing (stability fees) and loan demand. These findings are consistent with yield-seeking preferences or speculative motives within the cryptocurrency ecosystem (Cornelli et al., 2024).² Our findings also highlight the significant impact of protocol and blockchain-specific frictions. In Maker, loan demand is sensitive to the stability of its pegged-stablecoin, with greater volatility in stablecoin returns, resulting in a lower demand for debt creation. Moreover, consistent with prior research on the role of gas fees in decentralized trading (Barbon & Rinaldo, 2021; Capponi & Jia, 2021), we show a negative correlation between gas fees and borrowing activity, representing a unique cost component in this market setting.

Leveraging the granular, wallet-level transaction data available through blockchain technology, we next examine decentralized user leverage behavior within this leading collateralized borrowing protocol. Our unique setting allows us to directly test the impact of leverage on portfolio performance, contributing to a growing body of literature documenting underperformance among leveraged retail investors (Davydov & Peltomäki, 2023; Frazzini & Pedersen, 2022; Heimer & Imas, 2022; Subrahmanyam et al., 2024). Underlying this relationship are behavioral and structural mechanisms including, investors' tendency toward trend-chasing strategies (Barberis et al., 1998), pronounced disposition effects (Barber et al., 2020; Odean, 1999), and systematic overweighting of lottery-like securities with extreme return distributions (Hackethal et al., 2022; Kumar, 2009). Beyond these factors, Subrahmanyam et al. (2024) highlight forced liquidations triggered by margin calls as a key channel through which leverage exacerbates investor losses. The main cost incurred during such liquidations is the immediate price impact of the forced sale (or purchase), which can significantly erode portfolio value.

Whether leverage directly benefits or harms retail participants in cryptocurrency markets is an empirical question. Retail investor dominance in cryptocurrency markets coupled with minimal entry barriers creates conditions particularly conducive to leveraged speculation. However, the very features that make DeFi attractive also introduce substantial new risks, shifting the responsibility for due diligence and risk management to users BIS (2018). Cryptocurrencies used in DeFi transactions are highly volatile and, unlike traditional assets, are not backed by tangible assets or reliable income streams (Cheah & Fry, 2015; Yermack, 2015, pp. 31–43). Fundamentally, cryptocurrencies exhibit risk-return profiles that diverge markedly from traditional asset classes including equities, currencies, and commodities (Liu et al., 2022). They often display positively skewed return distributions (Liu et al., 2021), offering lottery-like payoffs compared to traditional markets (Zhao et al., 2024). This distinct return structure manifests in differential investor behavior that appeals to less-educated investors (Kumar, 2009), and without strong protections or regulatory oversight, they face heightened risks of financial loss due to limited understanding of the speculative nature of such assets. Kogan et al. (2024) document that investors who trade across both cryptocurrency and traditional equity markets employ contrarian strategies in stocks while adopting momentum-based approaches in cryptocurrencies.

Our detailed analysis of individual vault activity reveals a clear negative relationship between leverage and portfolio performance. We show that higher leverage lowers portfolio returns and increases the probability of liquidation. Vaults in higher leverage quintiles are younger, exhibit more transactional activity (both debt creation and repayment), lower average daily returns, and incur a higher incidence of liquidation events. We show that automated liquidations, an inherent feature of collateralized lending protocols designed

¹ On September 18, 2024, MakerDao officially rebranded as the Sky protocol, the Dai stablecoin was renamed USDS, and the MKR governance token became SKY. For consistency, we retain the original names throughout this paper.

² Whilst the core system of our analysis may not be prone to reflexive or incentive-driven behaviors, it is also not entirely immune from them. For example, leverage loops ('spirals'), represent a common leverage strategy, driven by the desire to amplify exposure to asset's price movements. Furthermore, the presence of external liquidity-minting protocols may indirectly stimulate credit demand within the Maker ecosystem by enabling users to recycle borrowed funds across platforms, thereby reinforcing leverage-driven borrowing behavior.

to maintain solvency, are a significant amplifier of this negative performance outcome. Analysis of liquidation events reveals that, on average, liquidation auctions result in collateral being sold at a discount of 7.6 % on a value-weighted-basis, and a substantial portion of the user's collateral (58.4 %) is sold to cover debt and penalties.

In addressing whether skill affects combined leverage and performance outcomes, recent research suggests a positive relationship between crypto and financial literacy, with better investment decisions resulting in greater wealth accumulation (Jones et al., 2024; Lusardi & Mitchell, 2023). Within decentralized markets, Aquilina et al. (2024a) show that a small group of participants exhibit considerable skill, extracting higher profits during periods of volatility by capturing a higher share of trading without incurring additional adverse selection. We follow Subrahmanyam et al. (2024) by undertaking a more nuanced examination of our leverage user group, assessing the dynamics between skill, leverage, and performance in the novel DeFi lending landscape. User skill is assessed on an out-of-sample, user-wallet basis, utilizing the on-chain verifiability of historical holdings. We show that skilled users manage moderate levels of leverage more effectively than their unskilled counterparts. However, excessive leverage proves detrimental to all users, with the negative consequences particularly severe for less skilled individuals. Our multivariate results reveal that in the highest leverage quintile, unskilled users experience an average daily performance decline of 40.1 basis points (bps), substantially larger than the 27.0 bps reduction observed among skilled users. These findings are partly driven by forced liquidations, however, for skilled users, a higher frequency of debt actions is positively correlated with performance, suggesting that active management of debt positions contributes positively to performance, an effect not observed among unskilled users. This distinction is critical, as it emphasizes that while DeFi platforms offer open and permissionless access to leverage, the ability to effectively manage that leverage is strongly linked to user sophistication.

Our findings collectively indicate that access to leverage presents inherent challenges related to cost and risk management that most participants find difficult to overcome profitably. Skilled users may be slightly better at avoiding liquidation events, particularly at moderate leverage. Nevertheless, the 'lottery-like' return characteristics of crypto assets used as DeFi collateral make pure leverage plays inherently riskier and more difficult to manage compared to leverage on traditional securities markets. Our results diverge from Subrahmanyam et al. (2024) where skilled futures traders benefit more directly from higher leverage. This result likely arises from several factors, including market structure and strategies, the nature of risks across each market, as well as definitions of skill. The authors present evidence that skilled investors earn rents from liquidity provision and basis arbitrage, strategies that not only benefit from scalability via leverage but are also less commonly associated with DeFi due to unique structural frictions, such as gas fees.

The findings of this research have several important implications for market participants. For DeFi users, the results emphasize the critical need for financial education, prudent risk management, and keen awareness of the severe negative consequences associated with excessive leverage risks to which even skilled participants are not immune. For protocol designers, the study highlights the necessity of incorporating advanced risk controls and user interface designs that better safeguard less experienced participants from the pitfalls of over-leveraging. Finally, for the broader financial ecosystem and regulatory bodies, this research contributes to a deeper understanding of risk dynamics and user behaviors in a rapidly evolving sector of the financial industry. As DeFi continues to grow in scale and complexity, empirical evidence on how users interact with its novel mechanisms is crucial for informed discussions regarding investor protection, market integrity, and financial stability.

2. Decentralized lending market

2.1. Defi lending protocols

DeFi lending protocols are a critical component of the broader DeFi ecosystem. Despite the myriad of financial, technological, and regulatory challenges facing the sector, the sector's rapid expansion over the past decade reflects growing market confidence in trustless financial infrastructure. Lending protocols account for approximately 25 % of over \$200 billion in TVL in DeFi, rivaling traditional margin markets in terms of scale. US stock-based loans, for example, total \$820 billion, roughly 2 % of the S&P 500's market capitalization, while margin lending in China constitutes around 10 % of domestic equity capital (Chen, 2019; Pang & Wang, 2020; Yardeni & Quintana, 2022).³

Within the DeFi ecosystem, access to decentralized lending is facilitated through a range of architecturally diverse protocols. These protocols rely on overcollateralization as a central mechanism to create leverage while mitigating counterparty risk arising from participant pseudonymity (Aramonte et al., 2022). Prominent protocols such as Aave, Compound, and Maker enable users to borrow against crypto-assets held as collateral. Typically, users pledge a higher-value asset to borrow a lower-value asset, creating a leveraged position. The borrowed asset can then be exchanged on decentralized markets for additional collateral, effectively establishing a recursive strategy that results in a leveraged long position in the collateral asset and a short position in the borrowed asset (Mueller, 2024). This approach enables users to amplify exposure while remaining within protocol-defined loan-to-value (LTV) constraints.

Borrowers can also utilize DeFi lending protocols to gain liquidity more conventionally. For example, Maker utilizes a vault-based system where users mint Dai stablecoins against collateral, which can be employed across DeFi platforms for trading, yield farming, and payments. This protocol provides fungible liquidity without selling their underlying collateral. However, the availability of this liquidity is closely tied to the value of the collateral, and as such, adverse price movements can significantly impair liquidity positions. IMF (2022) shows that the typical DeFi loan is disbursed in stablecoins, whereas the collateral is typically composed of riskier,

³ The Block, <https://www.theblock.co/data/decentralized-finance/total-value-locked-tvl>, accessed on January 4, 2025

unbacked cryptocurrencies.

Within this ecosystem, Maker, Compound, and Aave have emerged as dominant decentralized lending protocols.⁴ Although these protocols uniformly employ collateralized lending models due to limited borrower screening capabilities, they differ significantly in terms of collateral acceptance criteria, interest rate determination mechanisms, liquidation ratios and processes, and governance structures. For example, Compound is a decentralized lending protocol that utilizes a pool-based model, with interest rates dynamically adjusted based on utilization rates.⁵ Similarly, Aave employs a pool-based model but distinguishes itself through its unique offerings, such as flash loans (uncollateralized loans) that must be borrowed and repaid within the same transaction block. Aave's interest rate mechanism is segmented and highly sensitive to utilization rates. In contrast, Maker uses a vault-based model that allows users to generate Dai stablecoins against their collateral. Unlike the dynamically adjusted interest rate models employed by other lending protocols, Maker charges a governance-determined stability fee on user-generated debt. The stability fee serves as a lever to stabilize the peg mechanism associated with its stablecoin as opposed to a cost of capital charge used to balance the supply and demand in liquidity pools.⁶

Although collateralized borrowing and leverage are well-understood concepts in traditional financial markets, their implementation within decentralized blockchain markets introduces additional design frictions. The protocol design heterogeneity introduces complexities that present a unique setting for examining decentralized credit creation through algorithmic mechanisms. Unlike traditional systems where intermediaries manage credit risk, decentralized lending protocols transfer this risk to market-based liquidation processes. This shift introduces two key costs: the volatility of blockchain transaction fees and the reliance on third-party actors incentivized to execute liquidations. These costs disproportionately affect smaller traders, often pricing them out of leveraged positions during periods of high transaction fees (Mueller, 2024), whilst large-scale liquidations induce persistent price dislocations across decentralized exchanges (Lehar & Parlour, 2022). While over-collateralization serves as the primary credit protection mechanism (Chiu et al., 2024; Gadzinski et al., 2023), recent literature identifies potential systemic vulnerabilities arising from recursive borrowing patterns, where users employ borrowed tokens as additional collateral for subsequent loans (Campello et al., 2023). Comprehensive overviews of the general architecture of DeFi and its suite of protocols are provided by Schär (2021), Harvey et al. (2022) and Aspris et al. (2022).

2.2. The Maker protocol

The Maker protocol comprises a suite of smart contracts and technical infrastructure deployed on the Ethereum blockchain. These contacts integrate protocol, collateral, and vault-based management into a unified system governed by decentralized decision-making. Oversight is provided by MakerDAO, a decentralized autonomous organization (DAO) founded in 2014, which manages the growth, stability, and strategic direction of the Maker ecosystem. Fig. 1, illustrates time-series changes in collateral and debt positions for Maker from July 2019 to October 2023. Panel A presents the TVL in the Maker protocol and the amount of debt (Dai) borrowed through vaults. This panel captures the growth and subsequent decline in borrowing, closely mirroring broader trends in the cryptocurrency market. This is evidenced by the price of Ethereum (ETH) relative to the number of new vaults created on the platform over time shown in Panel B. The sharp rise observed between 2020 and 2022 shows the surge in adoption of cryptocurrencies and DeFi applications. Conversely, the post-2022 decline in borrowing reflects broader downturns in both traditional and crypto markets.

2.2.1. Multi-asset collateralization and vault management

Collateral is a central feature in many credit contracts (Benmelech & Bergman, 2009; Stulz & Johnson, 1985). It disciplines borrowers by allowing lenders to seize collateral in case of default. On the Maker protocol, collateral preserves the integrity and solvency of the Maker ecosystem. This approach operates through a permissionless yet governed model where MKR token holders vote to approve new asset types through governance proposals.⁷ Until November 2019, the Maker protocol supported only a single collateral (ETH). However, this framework was expanded to accommodate diverse collateral types, such as utility tokens, payment tokens, security tokens, stablecoins, and real-world assets (RWA) over time.⁸

2.2.2. Dai stablecoin

The operational integrity of the MakerDAO lending protocol depends fundamentally on the unique design of Dai as a decentralized stablecoin that maintains price stability. Dai preserves a stable value pegged to the USD through an overcollateralized debt mechanism, which forms the core structural foundation of the MakerDAO lending infrastructure. Users depositing ETH or other approved tokens

⁴ Centralized exchanges like Binance also offer leveraged trading products such as perpetual futures (De Blasis & Webb, 2022).

⁵ A comprehensive discussion of the Compound protocol can be found in Saengchote (2023); Lehar & Parlour (2022).

⁶ Dai is minted against collateral without requiring third-party liquidity, making the stability fee a unilateral cost rather than a market-determined price for capital. This structural difference eliminates the need for yield-bearing deposits, reframing the fee as a systemic maintenance cost rather than a competitive rate.

⁷ MKR token holders are the core decision-makers of the Maker protocol, using their tokens to vote on and implement changes that ensure the stability of the Dai stablecoin and the overall health of the ecosystem. For further discussion of DAOs and governance tokens see Han et al. (2025).

⁸ Appendix A lists the approved tokens eligible as collateral for Dai-denominated loans and outlines the detailed processes for onboarding new assets, including risk assessments, community governance votes, and technical integration. It also describes the offboarding procedures, which may be triggered by changes in market conditions, risk profiles, or governance decisions, ensuring the ongoing stability of the protocol.

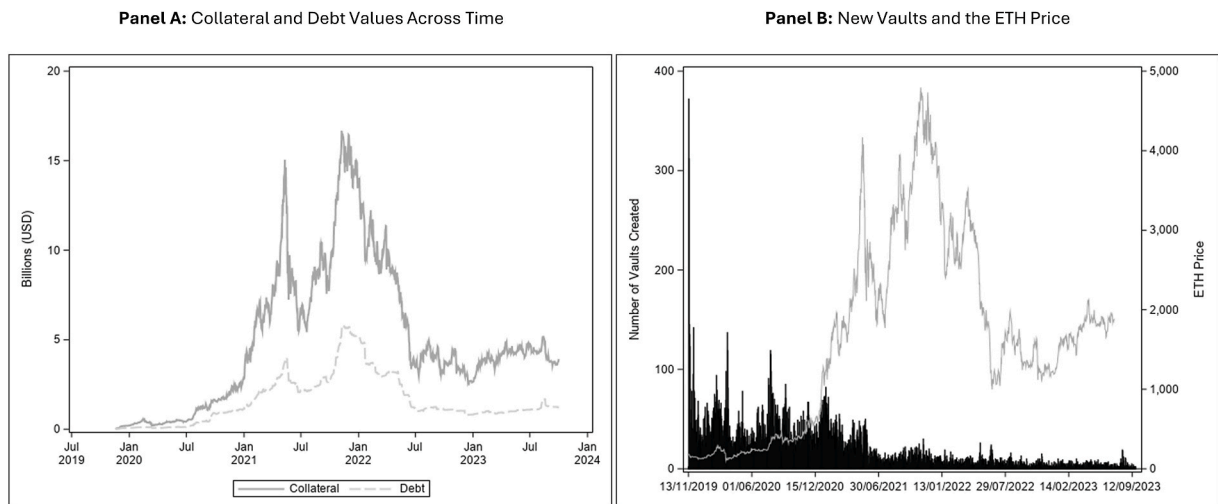


Fig. 1. Evolution of collateral and debt positions This figure tracks collateral and debt positions in Maker vaults over time. Panel A displays the value of locked collateral and debt origination within the Maker protocol from November 2019 to October 2023. Panel B shows the number of new vaults created, relative to ETH's market value. The analysis excludes loans from liquidity pools (e.g., CRVV1ETHSTETH-A), lending protocols (e.g., DIRECT-AAVEV2-Dai), and real-world assets (RWAs).

into Maker Vaults can mint Dai against their collateral, subject to predetermined collateralization ratios. The stability mechanism underpinning Dai directly facilitates the loan lifecycle through three interdependent components. These comprise collateralized debt positions (Section 2.2.3), programmatically enforced liquidations (Section 2.2.4), and governance mechanisms utilizing MKR tokens to dynamically calibrate risk parameters. This architectural framework ensures that loans denominated in Dai retain their nominal dollar value despite significant volatility in underlying collateral assets.⁹

2.2.3. Liquidation ratio, stability fee, debt ceiling

Operationally, debt issuance in the Maker protocol functions as a sophisticated form of collateralized borrowing, akin to a risky debt obligation. Users initiate this process by locking their approved crypto-assets into personalized Maker vaults, which are self-custodial and managed through Ethereum-based addresses. These vaults serve as the foundation for Dai creation, with the protocol minting new Dai tokens against the deposited collateral. The debt obligation is settled when borrowers repay their Dai-denominated loan, including accrued interest charges, to reclaim their collateral. Upon repayment, the Dai is burned and removed from circulation, maintaining the system's balance. The collateralization requirements for each collateral type are determined by the liquidation ratio, a critical risk parameter approved by MKR token holders for each collateral type. This ratio, influenced by factors such as the expected volatility of the collateral asset, establishes the level of overcollateralization necessary to avoid liquidation. For example, a liquidation ratio of 150 % mandates that the collateral value must consistently exceed 1.5 times the value of the generated Dai. If the collateral value falls below this threshold, the vault becomes subject to liquidation by the system, triggering a series of automated processes designed to protect the protocol's stability. This liquidation mechanism, crucial for maintaining the system's integrity, is explored in greater depth in Section 2.2.4.

A further key consideration in terms of debt issuance is the stability fee mechanism, which is a continuously compounded interest rate on Dai-denominated debt that serves as both an interest charge for borrowers and a risk management parameter. This fee is determined through decentralized governance by MKR token holders, who consider multiple factors, including Dai price stability and competitive yield conditions in global markets.¹⁰ Adjustments to the stability fee influence the cost of borrowing Dai. Specifically, higher fees are intended to discourage borrowing, thereby reducing the supply of Dai, whereas lower fees encourage borrowing, leading to an expansion in Dai supply.

The debt ceiling mechanism constitutes the third pillar of risk control, establishing collateral-type-specific issuance limits aligned with Maximum Exposure Levels (MELs). These MELs dynamically scale permissible Dai minting volumes relative to both collateral asset liquidity and systemic risk appetite. Governance maintains operational flexibility through ceiling adjustments that respond to market conditions, but binding ceilings serve as automatic circuit breakers. When reached, they immediately halt new Dai issuance against the affected collateral type, preventing concentrated risk accumulation.

A representation of the Maker balance sheet is illustrated in Fig. 2, highlighting the composition of assets, liabilities, and residual

⁹ The composability of Dai with Ethereum-based smart contracts facilitates its integration into DeFi lending markets, creating network effects that enhance liquidity within MakerDao. Consequently, borrowers can deploy borrowed Dai immediately into yield farming, liquidity provision, or risk-hedging strategies without with minimal price slippage.

¹⁰ Appendix B offers a detailed overview of the operational mechanics of the stability fee.

equity. The total value of outstanding loans currently stands at \$8.3 billion, with an average collateralization ratio of 157 %.¹¹ Other core features of the Maker protocol include savings, loans on RWA, and the facilitation of liquidity through its stablecoin.

2.2.4. Liquidations and auctions

The liquidation mechanism employed by the Maker protocol ensures systemic stability through a dual governance structure that combines predetermined risk parameters with economically incentivized third-party oversight. Liquidations are automatically triggered when a vault's collateralization ratio falls below the threshold specific to its collateral type. This framework prevents the accumulation of bad debt and enables swift, algorithmic enforcement of collateral requirements.

Central to this mechanism are keepers, automated agents analogous to high-frequency traders in traditional markets, who employ algorithmic monitoring systems to detect undercollateralized positions.¹² Keeper participation is incentivized through two coordinated mechanisms: firstly, a liquidation penalty, typically set at 13 % of the defaulted position, shared between keepers and the protocol, and secondly, discounted acquisition opportunities during collateral auctions. Total user losses therefore collectively include direct liquidation fees, the discount on auctioned collateral, and costs related to the auction actions, specifically gas fees. This structure creates a competitive environment where profit maximization aligns with timely debt resolution, as keeper profitability depends on rapidly executing liquidations ahead of competing agents.

Once liquidation is triggered, the protocol assumes the outstanding debt obligation and conducts a Dutch auction, characterized by an exponentially decaying price curve. This auction mechanism gradually lowers the collateral price until market demand clears the inventory, thereby fulfilling the debt obligation while minimizing price disruption through time-sensitive price discovery.¹³ The entire process—from threshold breach to final settlement—typically completes within minutes, demonstrating efficiency of the collateral recovery system. By integrating governance-defined risk parameters with market-based incentives for automated agents, the mechanism achieves self-regulating stability. The design of the protocol ensures that economic incentives directly reinforce systemic solvency requirements, while the auction mechanism guarantees final debt resolution regardless of collateral market conditions. A description of the liquidation process is provided in Fig. 3.

3. DeFi vs traditional finance

Financial innovation has long been intertwined with economic growth, often alongside technological advances (Laeven et al., 2015). Traditional finance relies on centralized entities to manage access, transactions and compliance. In contrast, DeFi is revolutionizing the financial sector by reducing the need for traditional intermediaries through the use of blockchain technology and smart contracts. This allows users to bypass regulatory barriers, granting direct access to financial services for individuals often excluded from conventional banking systems (Aquilina et al. (2024)). Users retain complete control over their assets, with all transactions and smart contract code transparently recorded on the blockchain.

However, the very features that make DeFi attractive also introduce substantial new risks, shifting the responsibility for due diligence and risk management to users (BIS, 2018). Cryptocurrencies used in DeFi transactions are highly volatile and, unlike traditional assets, are not backed by tangible assets or reliable income streams (Cheah & Fry, 2015; Yermack, 2015, pp. 31–43). These digital assets often display positively skewed return distributions (Liu et al., 2021), offering lottery-like payoffs compared to traditional markets (Zhao et al., 2024). Such characteristics appeal to less-educated investors (Kumar, 2009), and without strong protections or regulatory oversight, they face heightened risks of financial loss due to limited understanding of the speculative nature of such assets.

These risk are particularly pronounced in leveraged trading, which can magnify both gains and losses. Leveraged trading thus assumes a higher level of sophistication, including an understanding of market dynamics, risk management, and the mechanics of financial instruments. Paradoxically, despite its elevated risks such as margin calls, forced liquidations, and amplified losses, leveraged trading often appeals most to retail investors. This appeal stems from the promise of higher returns and amplified lottery-like payoff characteristics (Ladley et al., 2020).¹⁴ Bian et al. (2024) show that margin investors display significantly stronger preferences for these speculative payoffs compared to non-margin investors. They are less likely to possess firm-specific informational advantages and tend to be less sophisticated than short sellers (Chen et al., 2024). They also trade more frequently and exhibit poorer security selection ability compared to cash investors (Barber et al., 2020).

Our research into the behavior of leveraged traders contributes to several areas of research at the intersection of behavioral economics and finance. We begin our analysis by examining the dynamics of loan demand at the collateral level. Saengchote (2023) demonstrate that the demand for loans on the Compound protocol is positively related to borrowing rates, challenging conventional financial theory which posits a negative price elasticity of loan demand (see, Stiglitz & Weiss, 1981; Bernanke & Blinder, 1988; Bernanke & Gertler, 1995). He finds that, unlike traditional borrowers, DeFi users often borrow to engage in leveraged yield farming,

¹¹ As at 13 June 2024 (source: makerburn.com). ETH vaults have a current TVL of \$5bn with an average collateralization ratio of 551 %.

¹² Lehar and Parlour (2024) show that liquidation activity is highly concentrated, with the top 20 liquidators executing a significant portion of liquidations and collateral volume. This concentration points to a degree of specialization and suggests varying levels of efficiency among keepers.

¹³ If an auction fails to resolve, additional auctions may be conducted to sell excess Dai.

¹⁴ Collateral demands triggered by margin calls often lead to negative expected returns, yet positive skewness in the returns distribution.

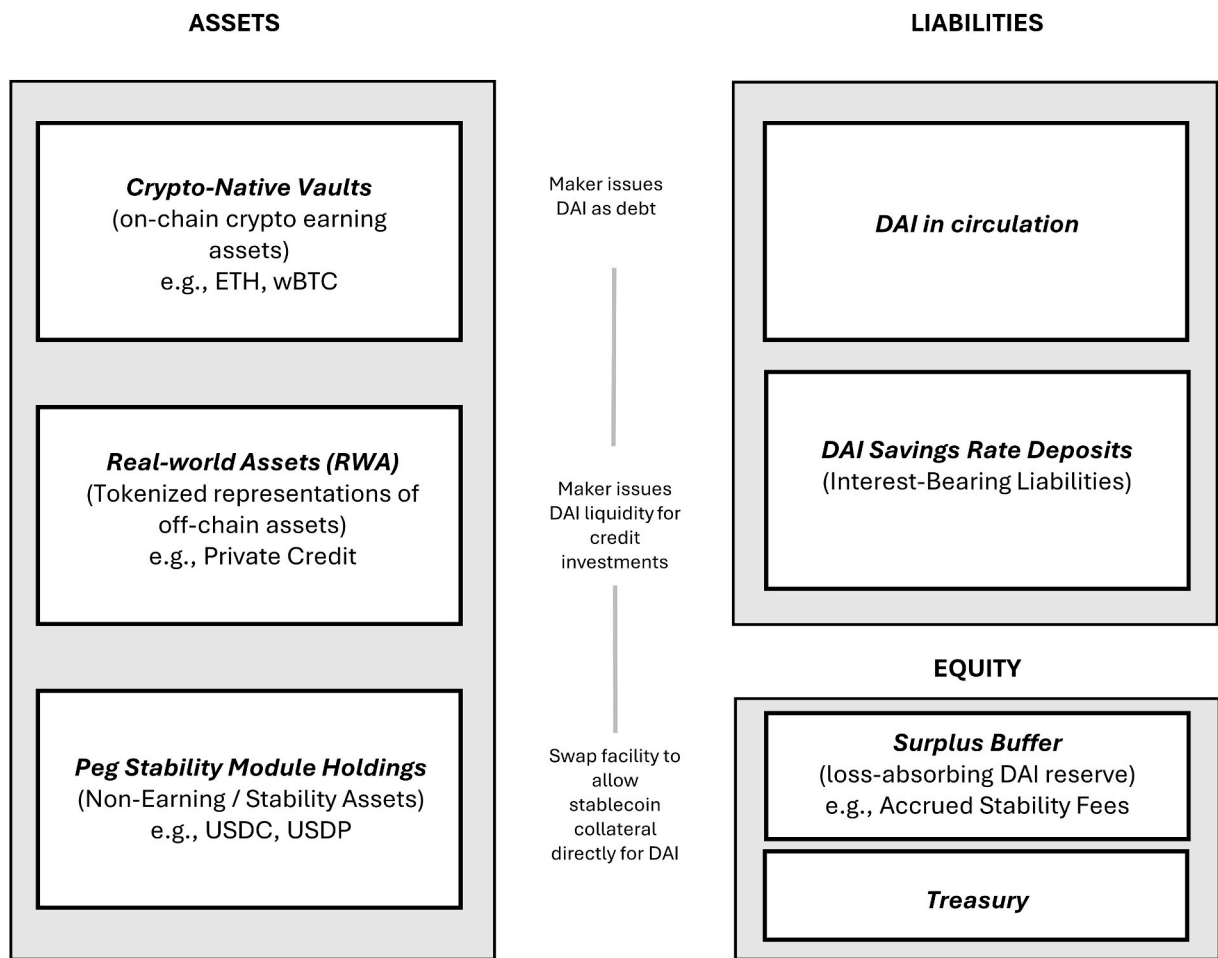


Fig. 2. MakerDao Balance Sheet This figure presents a stylized balance sheet of MakerDAO's core operations. Lending is automated via smart contracts, with users minting Dai by locking collateral in vaults. Upon repayment of the Dai loan and stability fees, borrowers reclaim their collateral. Stability fees contribute to the surplus buffer, a reserve of Dai accumulated over time. MakerDAO has expanded its collateral base to include off-chain real-world assets (RWAs), both private and public, which finance sectors like real estate, trade, and SMBs. Stablecoins are held in the Peg Stability Module (PSM), allowing fixed-rate swaps between Dai and other stablecoins to maintain Dai's peg. Dai holders can earn interest by locking tokens in the Dai Savings Rate (DSR) contract, funded by stability fees. Both the DSR and stability fee are dynamically adjusted based on deviations from Dai's target price. The surplus buffer is primarily sourced from stability fees and liquidation revenues, net of operational costs.

making them less sensitive to borrowing costs.¹⁵ COMP tokenomics therefore decouples loan demand from traditional rate sensitivity, as the demand for loans increases the utilization of token pools, causing interest rates to rise further. This was a fundamental novelty of DeFi protocols like Compound for a period after June 2020, where borrowers were incentivized to take loans despite higher lending rates because of the value of incentives (COMP governance token) awarded.

We revisit this hypothesis, noting that the Maker pool exhibits at least two institutional distinctions that are critical to understanding loan demand in DeFi markets. Notably, unlike other protocols, Maker does not employ a dynamically adjusted interest rate model based on token pool utilization. The stability fee (or interest charge), which is set by the governance team, serves as a lever to stabilize the Dai peg mechanism and the changes in this fee are less frequent. Consequently the relationship between interest rates and demand is less mechanical than on other protocols. Furthermore, unlike Compound, which uses the COMP token to incentivize borrowing and lending, MakerDAO does not implement a reward mechanism to stimulate user activity. The MKR governance token primarily facilitates governance decisions and recapitalizes the system in response to black swan events. Given these differences, it is unclear whether the positive relationship observed in [Saengchote \(2023\)](#) is intrinsic to DeFi lending market or protocol-specific, making it an open empirical question.

¹⁵ A simple example of a yield framing strategy involves user A borrowing token B at rate i_b , redepositing B into Compound to earn supply rate i_s and COMP rewards, yielding a net profit of $i_s + \text{COMP rewards} - i_b$; for example, borrowing USDC at 5 % and redepositing it to earn 3 % supply rate + 4 % COMP rewards generates a 2 % net gain even with $i_b > i_s$.

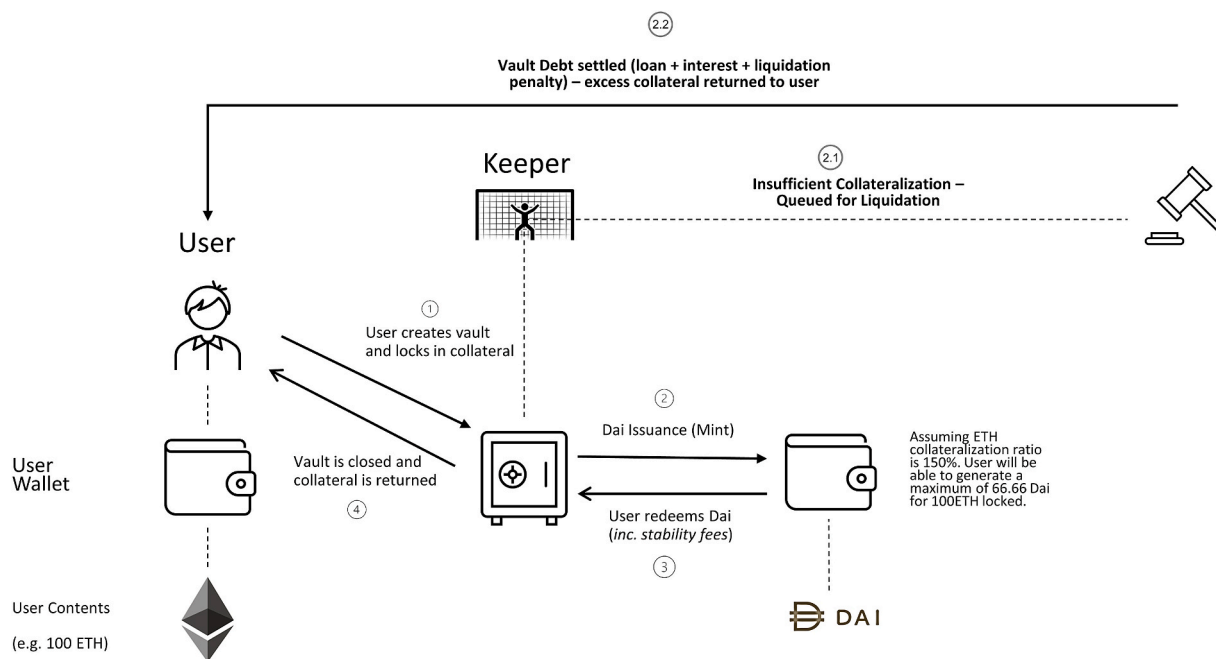


Fig. 3. Maker System This figure illustrates Maker's core lending mechanism. Borrowers mint Dai by locking collateral in vaults, forming a collateralized debt position. To repay, they return the Dai loan plus stability fees, reclaiming their collateral. The repaid Dai is burned, reducing supply. If collateral value drops below a set liquidation ratio, the protocol initiates liquidation via auction. The Dai raised covers the debt, with any surplus returned to the borrower.

Next, we investigate whether leverage in DeFi markets adversely affects performance, with particular attention to the role of forced liquidations. Research shows that leverage is generally detrimental to retail investor performance across various markets, including real estate (Tzur-Ilan, 2023), derivatives (Subrahmanyam et al., 2024), foreign exchange (Heimer & Imas, 2022; Heimer & Simsek, 2019), and equity markets (Ladley et al., 2020). However, regulators and brokers often impose strict leverage limits in these markets to discourage inexperienced and unskilled investors from overleveraging (Chen et al., 2024; Heimer & Simsek, 2019). In some jurisdictions, access to highly leveraged or complex financial instruments is further restricted to sophisticated investors who meet specific financial thresholds, reflecting heightened concerns about investor protection and product suitability. Similarly, banks use multitude of quantitative and qualitative data to guide decisions on user loan amounts, credit limits, and interest rates (Liberti & Petersen, 2019). This integrated approach enables more refined lending practices and tighter control over borrowing. In contrast, looser margin requirements in the DeFi market lead highly leveraged borrowers to increase their borrowing and further amplify their leverage (Wang, 2024). Moreover, liquidations on DeFi platforms are typically automated, offering users no opportunity to recapitalize their positions. Consequently, we hypothesize that looser borrowing constraints, relaxed leverage regulations, and forced liquidations lead to poorer outcomes for users.

Finally, we assess whether more sophisticated users outperform their less sophisticated counterparts and utilize leverage more efficiently. Subrahmanyam et al. (2024) show that leverage in futures markets negatively impacts performance due to increased trading costs and forced liquidations from margin calls. Nonetheless, they observe persistent performance differentials between skilled and unskilled investors, suggesting that user sophistication plays a critical role in navigating leveraged environments. However, considering the debated intrinsic value of digital assets (Cheah & Fry, 2015; Yermack, 2015, pp. 31–43) and the generally lower sophistication of participants in these markets, we hypothesize that the performance differential between skilled and unskilled users in DeFi markets will be less pronounced.

Our study differs from existing literature in two dimensions. First, research in traditional markets is often constrained by limited access to granular data, as individual retail trades and positions are typically unobservable or sourced from select brokerages, which may not represent the broader market. In contrast, the blockchain provides a transparent public ledger, capturing every transaction by users interacting with MakerDAO. This enables us to explore the effects of leverage on trading performance of retail investors in greater detail. Second, DeFi users exhibit a distinct risk profile from traditional markets. They borrow solely against posted collateral, lack investor protection, are exposed to forced liquidations, and face identical interest rates regardless of individual risk. To our knowledge, this is the first study to examine the interaction between traders' skill, leverage, and performance, and the mechanisms through which leverage impacts performance in DeFi markets.

4. Data

4.1. Data sources

The data used in this study comes from several sources. Primary data is acquired directly from the Maker protocol via its Application Programming Interface (API), spanning the period from November 13, 2019 to October 1, 2023. During this period, Dai ranked as the third-largest stablecoin, trailing only USDT and USDC, and was the largest decentralized stablecoin. The MakerDAO API retrieves data directly from the Ethereum blockchain, where all transactions and states are immutably recorded.¹⁶ The API provides detailed vault-level data, including comprehensive transaction history of individual users covering deposits, withdrawals, debt creation, debt payback, and liquidations.¹⁷ Each transaction is time-stamped to the nearest second and linked to a specific digital wallet addresses. The dataset comprises 31,061 unique vaults, encompassing 188,292 deposits, 139,999 withdrawals, 198,254 instances of debt generation, and 128,020 debt repayments. The total value of loans generated across these vaults exceeds \$33.3 billion, with over \$200 million in stability fees (interest) accrued. During the sample period, more than 4300 liquidation events were triggered to preserve the stability of the Maker protocol.¹⁸ Our compiled dataset also includes protocol-level parameters, detailing the collateral types onboarded and the specific risk parameters, such as the stability fee, liquidation ratio, and debt ceiling. We also capture governance-level voting statistics data at the protocol level, related to the onboarding of new collateral types.

We supplement our protocol-level panel data with several additional datasets. Trading data for all tokens used as collateral on Maker is sourced from [CoinMarketCap.com](https://www.coinmarketcap.com), a widely used resource in cryptocurrency research (Momtaz, 2021). This dataset includes daily observations on token prices, trading volume, circulating supply, and market capitalization. Furthermore, using vault wallet addresses identifier via the API, we collect on-chain daily token balance values of other ERC-20 crypto assets held by these wallets throughout the sample period. These token balances inform our definition of user skill, as detailed in Section 5.3.

4.2. Protocol level analysis

Table 1 presents the protocol summary statistics for Maker collateral over the sample period, categorized by each collateral risk tier. Panel A lists the crypto assets, while Panel B tabulates the stablecoins. Throughout the sample period, fifteen crypto assets and five stablecoins were onboarded onto the Maker protocol, allowing users to deposit their collateral and engage with the lending platform.¹⁹ The stability fee ranges from 0.24 % to 49.42 %²⁰, while the liquidation ratio varies from 1.01 to 3.50. Stablecoins have significantly lower liquidation ratios compared to crypto-asset-collateralized vaults, due to their lower volatility and risk profile, serving as a bridge between cryptocurrency and conventional financial systems. This flexibility in setting individual risk parameters controls the risk to the protocol from a drop in the price of a collateral asset. However, there is a trade-off: higher liquidation ratios make vaults less capital efficient, as they cannot issue as much debt, making them less attractive to users. More popular and relatively stable coins have a higher debt ceiling, which is the maximum amount of Dai that can be generated through collateralized debt positions. ETH, a leading base protocol coin for DeFi application, is the most popular form of collateral among vault users. Within specific collateral types, multiple tranches may exist which employ different combinations of protocol parameters. Notably, ETH collateral includes tranches such as ETH-A, ETH-B, and ETH-C, each offering distinct combinations of liquidation ratios and stability fees. ETH-A, the default ETH option, has a liquidation ratio of 1.47 and an average stability fee of 2.54 % over the observed period. In contrast, ETH-B maximizes Dai borrowing capacity per unit of ETH collateral at the expense of higher fees and increased risk exposure. It features a lower liquidation ratio of 1.30 and a higher stability fee of 4.95 %, enabling users with greater risk tolerance to generate more Dai, albeit at elevated costs and risks. Table 1 illustrates that the rapid growth in the overall value of collateral locked into the protocol is unevenly distributed across different collateral types. For instance, despite the Basic Attention Token (BAT) being part of the original launch of multi-collateral Dai, its contribution to the overall protocol has been negligible, with its collateral balance averaging only 0.42 % over the sample period. The lack of significant demand for this collateral type in Maker vaults has led to the offboarding of BAT, along with many other tokens including Compound (COMP), Uniswap (UNI), and 0x (ZRX).

¹⁶ The API data is retrieved from: <https://data-api.makerdao.network>. A web-based UI is available at <https://maker.blockanalitica.com/>. The API offers an efficient interface to access and interact with Ethereum blockchain data, which can be verified and further explored through Etherscan, a blockchain explorer.

¹⁷ The liquidation data includes details on the auction process, including auction rounds, degree of recovery, keeper participants, and price discount associated with the collateral auctioned.

¹⁸ This includes both partial and full liquidations conducted under the two distinct liquidation mechanisms (Liquidations 1.2 and Liquidations 2.0), that operated on the protocol during the sample period.

¹⁹ We exclude Uniswap and Curve LP tokens (e.g. UNI-V2-DAI-ETH and CRVV1ETHSTETH) which briefly appear over our sample period due to inherent structural differences and limitation utilization from our main analysis. We also exclude RWA vaults.

²⁰ While this number may seem high it is designed to compensate for regulatory and counterparty risks associated with USDC, which was issued by Circle, a centralized entity. It discourages excessive minting of Dai against USDC, which can increase systemic risks if USDC itself encounters liquidity issues.

Table 1

Maker Protocol Summary Statistics by Collateral Asset This table summarizes key operational and risk parameters for individual collateral assets within the Maker protocol. Panel A reports results for crypto-assets, while Panel B covers stablecoins, each over their respective debt origination periods. *# Days* indicates the number of days the collateral asset was actively used for debt origination during the sample period. *Stability Fee*, *Liquidation Ratio*, *Debt Ceiling*, and *Min Debt* represent, respectively, the median annualized stability fee (in percent), the minimum required collateralization ratio, the maximum allowable debt (in millions), and the minimum allowable vault debt. *Daily Return* and *Volatility* capture the average daily return and average daily price volatility (both in percent), with volatility calculated using the Parkinson method. These metrics are measured from the date the collateral was onboarded to the date of the last vault creation within the sample period.

	# Days	Stability Fee	Liquidation Ratio	Debt Ceiling	Min Debt	Daily Return	Volatility
Panel A: Crypto-Assets							
AAVE-A	336	2.73	1.72	6.93	4004.46	0.62	6.74
BAL-A	374	2.26	1.73	3.15	5466.58	0.38	6.56
BAT-A	693	3.93	1.50	4.37	2401.56	0.34	5.24
COMP-A	418	2.26	1.75	4.31	4962.92	0.46	6.69
ETH-A	1419	2.54	1.47	897.25	8586.60	0.27	3.51
ETH-B	1078	4.95	1.30	74.35	23,200.93	0.24	3.58
ETH-C	927	1.16	1.71	409.85	4606.80	0.09	3.37
GNO-A	165	4.98	3.50	4.98	100,000.00	0.20	1.98
KNC-A	331	2.57	1.75	3.97	1404.29	0.38	6.87
LINK-A	945	2.48	1.69	34.78	11,046.56	0.14	5.07
LRC-A	376	3.44	1.75	2.45	4400.27	0.54	7.35
MANA-A	843	6.02	1.75	5.57	6748.78	0.64	6.46
MATIC-A	599	3.00	1.75	18.33	14,257.10	0.12	4.62
RENBTC-A	699	3.28	1.69	4.13	9367.67	0.04	3.27
RETH-A	315	1.85	1.65	14.51	10,595.24	0.18	1.94
UNI-A	541	2.42	1.70	12.23	7500.92	0.32	6.14
WBTC-A	1246	3.12	1.47	301.53	8373.53	0.15	2.71
WBTC-B	676	4.72	1.30	31.09	28,276.63	−0.06	2.09
WBTC-C	672	1.88	1.75	106.22	6113.10	−0.06	2.09
WSTETH-A	703	2.42	1.59	144.00	12,222.62	−0.05	2.92
WSTETH-B	504	1.16	1.84	174.37	4306.55	0.05	2.75
YFI-A	902	2.33	1.68	20.74	8109.31	0.14	5.18
ZRX-A	470	3.10	1.75	4.34	3531.53	0.52	6.59
Median	636	2.56	1.70	16.42	7124.85	0.19	4.10
Panel B: Stablecoins							
GUSD-A	420	0.24	1.01	5.00	6966.67	0.00	1.19
PAXUSD-A	256	1.41	1.01	94.02	1802.34	0.00	0.47
TUSD-A	237	1.87	1.10	71.73	182.70	−0.00	0.33
USDC-A	352	3.30	1.10	264.84	332.05	0.01	0.48
USDC-B	355	49.42	1.20	26.85	1238.59	0.00	0.24
USDT-A	231	7.93	1.50	4.55	1435.07	0.00	0.15
Median	352	3.30	1.01	26.85	1238.59	0.00	0.33

4.3. Collateral and vault summary statistics

Table 2 presents summary statistics for the comprehensive set of vault records in our sample period across all crypto-asset collateral risk tiers. Panel A reports aggregated statistics for each collateral risk-tier variant, providing an overview of the collateral loan dynamics for the protocol. Panel B on the other hand, summarizes individual vault behavior independent of the chosen collateral risk tier. From this sample, all stablecoin, RWA, and LP vaults are excluded because these underlying tokens have fundamentally different underlying characteristics and operate under distinct risk parameterization frameworks and centralized asset dependencies. This exclusion criteria ensures the dataset reflects only vaults collateralized by native blockchain assets (e.g., ETH, WBTC) governed by MakerDAO's core decentralized risk models, rather than those influenced by external price stabilization mechanisms (stablecoins), RWA custodial risks, or liquidity pool impermanent loss factors (LP positions). We also remove vaults from the total sample that either did not generate any debt or were active for fewer than ten days.

Panel A reveals a substantial amount of collateral added and debt generated for the average collateral variant on Maker, with a mean value of \$2.3 billion and \$1.1 billion, respectively. The standard deviations of these values, however, highlight considerable variability in the choice of collateral form. Similarly, average collateral and debt actions, show frequent vault interactions, indicating an active user base that manages their positions to either optimize their returns or mitigate risks in response to market movements. However, the disparity between the upper and lower percentiles reveals a considerable preference for particular collateral forms. These findings elucidate the decision of the governance team to remove a substantial portion of the collateral variants listed in **Table 1** during the sample period, as the insufficient demand did not justify the associated costs. Liquidation values and actions are heavily right-skewed which is consistent with the fact that a high proportion of vaults are concentrated in few collateral forms.

Panel B provides a granular view of individual vault characteristics, revealing fundamental insights about user behavior within the Maker protocol. The mean vault age of 0.88 years with a standard deviation of 0.85 years indicates substantial variability in user

Table 2

Collateral and Vault Level Statistics This table presents summary statistics for vault activity across all crypto-asset collateral types. Panel A reports data aggregated at the collateral-type level, while Panel B summarizes statistics at the vault-day level. In Panel A, *Collateral Added (Withdrawn)* is the total value of collateral added (withdrawn). *Debt Added (Payback)* is the total value of Dai generated (repaid). *Collateral Actions* count the number of collateral-related transactions, including deposits and withdrawals. *Debt Actions* indicate the number debt-related transactions, including debt generation and repayments. *Liquidation Actions* indicate the number of liquidation events, while *Liquidations* reflect the value of these liquidations. All monetary amounts are expressed in millions. Panel B includes additional vault-level characteristics. *Vault Age* is the age of the vault in years. *Collateralization ratio* is the ratio of collateral value to cumulative debt generated. *Collateral* represents the total value of locked collateral. *Debt Value* is the daily outstanding debt. *Cumulative return* is calculated as the percentage difference between capital withdrawn and invested, relative to invested capital. *Stability Fees* denote the value of fees charged to the vaults. Daily collateral and debt values are reported in millions, while collateral and debt transaction values are expressed in thousands.

	n	Mean	Std	p5	p25	p50	p75	p95
Panel A: Collateral-level Analysis								
Collateral Added	23	2394.68	5673.48	10.66	35.48	162.62	2683.59	9250.78
Collateral Withdrawn	23	2272.34	6181.44	4.57	29.35	142.98	1310.21	8134.35
Debt Added	23	1104.88	2850.00	3.10	14.26	46.23	930.30	3862.14
Debt Payback	23	1027.20	2754.27	1.65	13.26	27.65	459.82	3750.55
Collateral Actions	23	13,653.39	46,092.50	152	356	1381	5402	24,671
Debt Actions	23	13,632.65	45,558.32	184	357	1510	5314	28,162
Liquidation Actions	22	389,091	1355.16	1	12	31.5	124	502
Liquidation Value	22	19.00	45.82	0.13	0.18	1.48	7.26	77.40
Panel B: Vault-level Analysis								
Vault Age	6,019,196	0.88	0.85	0.04	0.22	0.60	1.29	2.70
Collateralization Ratio	4,835,595	10.02	113.71	1.94	2.61	3.68	6.67	33.77
Collateral Value	6,019,196	1132.09	22,697.66	0.00	0.82	20.16	131.88	1717.66
Collateral Added	133,877	411.41	6711.95	0.10	1.72	10.14	51.36	630.66
Collateral Withdrawn	86,605	603.47	9649.55	0.13	3.07	17.81	90.98	1087.86
Collateral Actions	201,306	1.56	1.71	1.00	1.00	1.00	2.00	4.00
Debt Value	6,019,196	334.13	7810.49	0.00	0.02	4.44	34.87	466.23
Debt Added	150,802	168.52	2067.28	0.07	1.20	7.00	30.47	350.17
Debt Payback	90,346	261.50	3051.69	0.06	1.84	10.50	50.67	576.41
Debt Actions	221,427	1.42	1.39	1.00	1.00	1.00	1.00	3.00
Liquidation Actions	4322	1.98	14.76	1.00	1.00	1.00	1.00	3.00
Liquidations Value	4322	96.71	1047.92	0.02	0.20	5.00	22.58	246.69
Cumulative Return	26,422	59.00	217.85	-46.74	-9.08	3.36	38.04	350.89
Stability Fees	26,422	6347.09	212,902.50	0.01	1.73	36.06	373.05	7137.59

commitment timeframes, with most vaults being relatively young (median of 0.22 years) while a small percentage exceed 2.7 years in age. The average collateral value is approximately \$1.13 million, with a mean debt exposure of \$334,130. Whilst this suggests significant capital commitment from the average user, the results are positively skewed. The median collateral value for each vault is \$20,160 with a median debt value of \$4440 which shows that the typical vault is relatively modest in size, with a smaller number of exceptionally large vaults. The median collateralization ratio of 3.68 suggests substantial collateralization within the protocol, serving as a buffer against market volatility at the expense of capital efficiency for users. In DeFi lending, higher collateralization ratios mean borrowers must lock up more collateral relative to their loan amount, increasing the opportunity cost for borrowers. Our activity metrics at the vault level reveal conservative management practices among users. With means of 1.56 collateral actions and 1.42 debt actions, most users make infrequent adjustments to their positions. Finally, the cumulative profit metric provides crucial insight into user performance within the protocol. We define performance as the level of return on user invested capital over a full deposit-withdrawal cycle for a vault, and show a mean cumulative return on capital of 59 %. Once again, there is significant variability in this figure, with the 5th percentile showing losses of 46.74 % and 95th percentile showing gains exceeding 350.89 %. This highly dispersed return profile illustrates the varied leverage strategies, timing decisions, and outcomes within the protocol.

5. Empirical analysis

5.1. Loan demand

To examine the factors affecting loan demand (*LD*) within a multi-collateral framework, we specify the following panel regression:

$$LD_{it} = \beta_0 + \sum_{p=1}^P \beta_{1p} \text{Protocol}_{p,it-1} + \sum_{c=1}^C \beta_{2c} \text{Collateral}_{c,it-1} + \sum_{o=1}^O \beta_{3o} \text{Other}_{o,it-1} + \alpha_i + \gamma_t + \epsilon_{it} \quad (1)$$

where, LD_{it} represents loan demand associated with collateral i at day t , for four specifications of loan demand. First, we measure loan demand as the cumulative value of outstanding debt positions. Second, we express loan demand as the daily amount of new loan positions created on collateral i at time t . Third, we consider the net-value of loans generated (loans generated less loans repaid) and

finally, we examine loan utilization measured as the cumulative outstanding loan balances relative to the hypothetical debt ceiling. The hypothetical debt ceiling is the value of collateral locked into the protocol divided by the collateral liquidation ratio. In the model, β_0 is the intercept while $\sum \beta_{1p} \text{Protocol}_{p,i,t-1}$ represents the protocol parameters, which include the stability fee and the liquidation ratio. The term $\sum \beta_{2c} \text{Collateral}_{c,i,t-1}$ includes collateral-related factors such as market volatility, collateral returns, and the age of vaults associated with the collateral. The term $\sum \beta_{3o} \text{Other}_{o,i,t-1}$ captures other significant drivers including average gas fees for transacting on the protocol, the liquidation spread, defined as the absolute difference between the aggregate collateralization ratio, the liquidation ratio, and the peg, which is the daily volatility of the Dai stablecoin and reflects the financial stability of the protocol. All model specifications include collateral-tier and time-specific fixed effects, controlling for changes in loan effects across different collateral types and temporal periods. Finally, ϵ_{it} is the error term. Standard errors are clustered at the collateral and date levels.

Table 3 presents the regression results, highlighting several significant determinants of loan demand within the protocol. Notably, the stability fee (interest rate) coefficient is positive and statistically significant across multiple model specifications. This indicates that higher interest rates are associated with increased loan demand, consistent with the findings of Saengchote (2023). This suggests that, despite institutional differences, the result may be inherent to decentralized lending markets. Several mechanisms could explain this divergence from conventional financial theory. Notably, even in the absence of explicit reward structures within the Maker protocol, yield-seeking behavior and overconfidence may drive borrowers to accept higher interest rates in pursuit of potentially greater returns elsewhere, thereby preserving positive expected arbitrage margins. This aligns with established behavioral finance theories that explain speculative trading among less sophisticated investors (Heimer & Simsek, 2019). Additionally, the result may reflect a willingness among borrowers to incur higher costs in exchange for immediate liquidity.

The other key protocol parameter, the liquidation ratio, exhibits a strong negative relationship with loan demand across all model specifications. This indicates that stricter collateral requirements discourage borrowing and reduce utilization rates. This finding is in line with economic intuition that higher collateralization requirements constrain borrowing capacity. The result aligns with Tzur-Ilan (2023), who show that stricter LTV limits reduce mortgage loan amounts in the real estate market. It is also consistent with Heimbach and Huang (2024), who demonstrate that users utilize all available leverage and exhibit higher gearing when offered a higher LTV ratio.

The regression results highlight the importance of blockchain-specific frictions in DeFi lending markets. In particular, gas fees exhibit a significant negative association with borrowing activity, reflecting a unique cost component absent in traditional financial intermediation. This finding is consistent with prior research, which shows that gas price fluctuations reduce trading volumes on

Table 3

Loan Demand This table reports the results of aggregate loan demand regressions over the sample period, using four distinct measures of loan demand. *Debt* refers to the cumulative outstanding Dai debt on the Maker protocol for each collateral type. *Debt Generated* captures the daily amount of newly issued debt, while *Net Debt Generated* reflects the net issuance, calculated as debt generated minus debt repaid. *Utilization* ratio is defined as the ratio of cumulative outstanding Dai to the hypothetical debt ceiling, which is derived by dividing the total collateral locked in the protocol by its liquidation ratio. The explanatory variables include *Stability Fee*, which is the interest rate charged on the debt generated for a given collateral. *Liquidation Ratio* is the minimum liquidation ratio set by the protocol. *Collateral Added* is the amount of collateral deposited and locked in the protocol, denominated in Dai. *Age* indicates the average age of vaults associated with the collateral, measured in years. *Return (Collateral)* and *Volatility (Collateral)* are the daily return and volatility of the underlying collateral. *Gas* is the average daily gas cost. *Liquidation Spread* is the absolute difference between the aggregate collateralization ratio and the liquidation ratio, while *Peg* is the daily volatility of the Dai stablecoin. All explanatory variables are lagged by one day. Statistical significance at the 10 %, 5 %, and 1 % level is indicated by***, ***, and *, respectively.

VARIABLES	Debt	Debt Generated	Net Debt Generated	Utilization
Stability Fee	3726* (1993)	41.29* (22.79)	-6.534 (9.419)	1.571*** (0.546)
Liquidation Ratio	-1239* (708.4)	-17.52** (8.299)	-0.484 (4.927)	-0.540** (0.242)
Collateral Added		0.0557*** (0.0104)	0.00923 (0.0162)	
Age	-892.5*** (274.3)	-10.59*** (3.325)	-0.938 (0.759)	-0.0302 (0.0593)
Return (Collateral)	28.96 (69.68)	-1.782 (3.162)	-2.059 (5.610)	-0.101 (0.0838)
Volatility (Collateral)	-4.651 (87.82)	4.245 (4.203)	4.644 (5.696)	-0.134* (0.0649)
Gas		-0.157** (0.0745)	-0.110* (0.0550)	-0.00689*** (0.000617)
Liquidation Spread	-0.0135 (0.778)	0.0175 (0.0115)	0.0257 (0.0210)	-0.00412* (0.00239)
Peg	-2.719 (2.862)	-0.191*** (0.0228)	-0.180*** (0.0138)	-0.00257*** (0.000380)
Constant	2130* (1165)	45.22*** (13.84)	7.540 (9.451)	1.994*** (0.415)
Observations	15,232	15,232	15,232	15,232
R-squared	0.644	0.285	0.108	0.633
Vault Fixed Effects	Yes	Yes	Yes	Yes
Day Fixed Effects	Yes	Yes	Yes	Yes

decentralized exchanges (Makarov & Schoar, 2022), and that higher transaction fees decrease the daily propensity to rebalance (Mueller, 2024). The stability of the Dai peg also emerges as a significant determinant of loan demand and underscores the importance of stablecoin reliability in facilitating lending activity within decentralized protocols.

Lastly, the addition of collateral is positively associated with loan generation, though this effect is only significant in the Debt Generation specification. This suggests that users who lock in more collateral are likely to generate additional debt. In contrast, the aggregate age of vaults shows a significant negative relationship with loan demand, indicating that older vaults are less active in borrowing. This may reflect behavioral tendencies such as declining user engagement over time or differing risk preferences between newer and more established users. Collectively, these findings enhance our understanding of DeFi lending behavior, emphasizing the influence of both economic incentives and behavioral dynamics that diverge from those observed in traditional financial systems.

5.2. Leverage and vault performance

In this section, we analyze the effect of leverage on vault performance. Extensive literature shows that leverage tends to harm retail investors' performance in traditional financial markets (See, for example, Subrahmanyam et al., 2024; Tzur-Ilan, 2023). Building on prior research, we investigate whether greater user exposure to collateral through leverage trading enhances or diminishes user performance on average, and whether some users can earn excess returns by employing higher leverage. In line with Subrahmanyam et al. (2024), we consider whether skill, assessed out-of-sample, drives differences in relative performance, particularly through the use of leverage and the occurrence of forced liquidations resulting from undercollateralized positions.

We assess user skill based on vault performance over the 100 trading days preceding the creation of their Maker vault. Each vault is linked to a corresponding wallet address, identified via the Maker API and verified on-chain. Using this linkage, we retrieve historical token balances for the pre-vault period. Daily portfolio returns are computed as the percentage change in the total value of holdings and benchmarked against daily Bitcoin price movements to assess relative performance using a simple market proxy. Based on cumulative excess returns, we categorize users into three groups: the top 30 % are classified as 'skilled', while the bottom 30 % are labeled 'unskilled'. These classifications are subsequently used to analyze vault performance within the Maker ecosystem.

While our proxy for skill is intuitively appealing, it has several limitations. Benchmarking against Bitcoin provides a parsimonious market proxy, but it overlooks exposure to broader systematic risk factors such as size and momentum, that are empirically validated drivers of cross-section returns (Liu et al., 2022). By omitting these dimensions, our classification of excess returns may conflate true investment skill with compensation for latent factor exposures.²¹ Beyond benchmarking limitations, we cannot ascertain whether the wallet linked to the Maker protocol is the only wallet belonging to the user or part of a larger portfolio that includes non-ERC-20 tokens. Consequently, our skill metric captures observed performance rather than comprehensive investing ability.²² Nevertheless, this approach offers key advantages for studying performance in DeFi. Temporal precedence and on-chain verifiability ensure skill measurement precedes vault outcomes, creating an auditable performance record while avoiding the selection biases and opacity common in conventional performance studies.

To address the performance of user vaults within the Maker protocol, we first establish a framework for measuring the economic returns generated by each vault over its operational lifetime. This approach ensures our performance metrics accurately capture the dynamic relationship between collateral management, debt issuance strategies, and user-driven cash flows. The economic value of vault (v) on day t is its Net Asset Value ($NAV_{v,t}$), defined as:

$$NAV_{v,t} = CV_{v,t} - D_{v,t} \quad (2)$$

where $CV_{v,t}$ is the market value of the collateral locked within vault v at time t , and $D_{v,t}$ is the total outstanding debt obligation of vault v at time t , inclusive of any accrued stability fees as detailed in Appendix B. We next define the net cash flow to the user from the vault during period t :

$$NCF_{v,t} = (DG_{v,t} + CW_{v,t}) - (CA_{v,t} + DR_{v,t}) \quad (3)$$

where $CA_{v,t}$ is the value of additional collateral deposited into vault v by the user, $CW_{v,t}$ is the value of collateral withdrawn from vault v by the user, $DG_{v,t}$ is the value of Dai generated (borrowed) by the user from vault v , and $DR_{v,t}$ is the value of Dai repaid by the user to vault v . A positive $NCF_{v,t}$ indicates a net withdrawal of funds by the user from the vault during period t , while a negative value signifies a net injection of capital.

The cumulative profit of vault v up to time T ($CP_{v,T}$), representing the total economic gain to the user from the vault's inception (T_0) to time T , is then defined as the sum of the vault's current Net Asset Value and the sum of all net cash flows extracted by the user over its lifetime:

$$CP_{v,T} = NAV_{v,T} + \sum_{t=T_0+1}^T NCF_{v,t} \quad (4)$$

²¹ Unlike equity markets, where standardized factor datasets enable systematic risk decomposition, crypto asset markets lack consensus benchmarks for size, momentum, or volatility factors. The absence of institutional-grade historical databases tracking token-level characteristics makes multi-factor adjustments infeasible.

²² This is often a constraint faced by empiricists studying frontier markets.

This formulation captures both the change in the vault's intrinsic value and the net inflows or outflows experienced by the user over time. To assess daily performance, we calculate the daily Holding Period Return (HPR) for each vault v for day t , denoted as $R_{v,t}$. The profit for day t , representing the change in cumulative profit ($\Delta CP_{v,t}$), is given by:

$$R_{v,t} = \frac{\Delta CP_{v,t}}{NAV_{v,t-1}} = \frac{(NAV_{v,t} - NAV_{v,t-1}) + NCF_{v,t}}{NAV_{v,t-1}} \quad (5)$$

which accounts for both changes in market valuation and net user transfers. This formalized framework, employing NAV and HPR, captures the key drivers of vault profitability, including changes in collateral valuation, the cost of debt (stability fees), and the impact of user-led active debt and collateral management strategies.

Table 4 summarizes the performance statistics at the vault level. Panel A presents the complete dataset segmented by leverage quintiles, with average leverage increasing from 15.3 % in the first quintile to 34.90 % in the fifth. Leverage is defined as the inverse of the collateralization ratio, offering a more intuitive measure of the debt generated relative to the collateral held within the vault. The table indicates that higher vault leverage is common in younger vaults and is associated with increased transactional activity, including both debt creation and repayment. Higher leverage is also negatively correlated with daily returns and is associated with a higher probability of vault liquidation, suggesting that liquidations may be contributing to negative performance.

Panels B and C present the results of our user skill analysis, showing that unskilled investors, on average, maintain younger vaults, achieve lower daily returns, and experience an incrementally higher rate of liquidations compared to skilled users. The daily return metrics highlight a significant economic difference between the two user groups. Although both exhibit declining returns as leverage increases, skilled users maintain positive returns even within the highest LTV group (0.108), whereas returns of unskilled users fall below zero (−0.061). This performance gap widens significantly at higher leverage levels, with the high-low difference being much more pronounced for unskilled users (−0.530) compared to skilled users (−0.330). These findings indicate that expertise helps to mitigate the negative impact of leverage on returns, allowing more knowledgeable participants to manage the risks of higher leverage more effectively. Interestingly, unskilled users perform more vault actions in the highest LTV group (0.207) compared to skilled users (0.181), possibly indicating reactive rather than strategic behavior during periods of market stress. Despite increased activity, performance does not improve, as reflected in the liquidation rates. Both user groups experience increasing liquidation frequencies with

Table 4

Vault Level Descriptives This table presents vault level descriptives. Panel A presents descriptive statistics on vault leverage, performance, and operational characteristics, segmented by leverage quintile. *Vault Leverage* is the average LTV ratio, calculated as the ratio of generated Dai to the value of locked collateral. *Vault Age* measures the lifespan of a vault in years. *Vault Actions* indicates the average number of daily on-chain debt transactions over the vault's lifetime (including both Dai generation and Dai repayment). *Daily Return* measures the daily percentage change in the vault's net value (collateral value minus outstanding Dai debt). *Liquidations* is a binary indicator equal to one if the vault experienced an automated, auction-based liquidation within the Maker protocol, zero otherwise. Panels B and C present identical metrics for users classified as skilled and unskilled, respectively. User skill is determined by ex-ante abnormal performance (returns adjusted for market factors) derived from their blockchain portfolio data during a 'clean period' prior to vault creation. Skilled (unskilled) users are those in the top (bottom) 30 % of abnormal returns during this period. The bottom rows of each panel report the differences between the highest and lowest quintiles, along with p-values testing the null hypothesis of no significant difference.

LTV Group	Vault Leverage	Vault Age	Vault Actions	Daily Return	Liquidations
Panel A: Combined					
Low	0.153	1.022	0.032	0.454	0.051
2	0.217	0.852	0.050	0.367	0.106
3	0.252	0.783	0.062	0.310	0.150
4	0.286	0.657	0.089	0.192	0.219
High	0.349	0.661	0.194	0.018	0.325
High - Low	0.196	−0.362	0.163	−0.436	0.274
p-Value	<0.001	<0.001	<0.001	<0.001	<0.001
Panel B: Skilled users					
Low	0.153	1.153	0.030	0.438	0.047
2	0.216	0.904	0.050	0.395	0.109
3	0.252	0.861	0.065	0.317	0.167
4	0.286	0.714	0.082	0.241	0.218
High	0.353	0.719	0.180	0.108	0.315
High - Low	0.199	−0.434	0.151	−0.330	0.267
p-Value	<0.001	<0.001	<0.001	<0.001	<0.001
Panel C: Unskilled users					
Low	0.153	0.906	0.033	0.469	0.053
2	0.217	0.801	0.049	0.339	0.103
3	0.252	0.707	0.058	0.303	0.134
4	0.285	0.600	0.096	0.143	0.219
High	0.346	0.609	0.207	−0.061	0.333
High - Low	0.193	−0.297	0.174	−0.530	0.280
p-Value	<0.001	<0.001	<0.001	<0.001	<0.001

higher LTV ratios; however, unskilled users consistently face slightly higher rates in the most leveraged category (0.333 vs 0.315). The persistence of this pattern across all LTV quintiles highlights the critical role of user skill in managing risk and preserving capital within DeFi protocols.

To explore the economic implications of liquidation costs, Table 5 presents descriptive statistics across different collateral tiers, highlighting key patterns in liquidation behavior and market responses. It shows a pronounced concentration of liquidation events among Ethereum-based collateral types, reflecting Ethereum's dominant position within the DeFi ecosystem. ETH-A collateral exhibits the highest number of liquidations, totaling 3017 events, while contributing approximately 49 % of total debt generation (\$13 billion). Despite this significant debt issuance, ETH-A maintained a relatively low liquidation ratio of around 1.5 % (\$201 million liquidated), indicating strong collateral stability within this tier. Conversely, alternative collateral types such as LINK-A demonstrate higher vulnerability to liquidation events, with 3.2 % of their generated debt being liquidated. The auction process efficiency, as measured by price discounts relative to fair value, exhibits significant heterogeneity across collateral types and transaction sizes. Equal-weighted (EW) discounts, which assign equal importance to all liquidation events, average 2.3 % across collateral types, while value-weighted (VW) discounts, which place greater emphasis on larger liquidations, a significantly higher 7.6 %. This pronounced size effect, where larger liquidations exert greater price impact during auctions, is likely driven by limited market depth, liquidity constraints, or strategic bidding behavior by participants. Collateral-specific discount patterns further illuminate underlying market microstructure variations. BAL-A (EW: 3.0 %, VW: 2.0 %) and YFI-A (EW: 2.6 %, VW: 1.1 %) demonstrate distinct price discovery dynamics compared to more liquid collateral types such as ETH-A (EW: 2.2 %, VW: 8.7 %). These disparities suggest differing liquidity premia and information asymmetries within the cryptocurrency asset market.

A key aspect of the liquidation process is the proportion of collateral liquidated to cover outstanding debt and associated fees. The results show that on average, 73.2 % of collateral is liquidated on an equal-weighted basis, leaving approximately 26.8 % returned to vault owners post-liquidation. However, when weighted by transaction value, the average proportion decreases to 58.4 %, suggesting more favorable collateral retention outcomes for larger positions. These findings underscore the systemic resilience of collateralized lending protocols. With total debt generation amounting to \$27 billion and only \$679 million liquidated (2.48 %), the data provide strong evidence of effective risk management practices, even amid market volatility.

Fig. 4 illustrates the time-series evolution of liquidation activity from November 2019 to October 2023. Panel A highlights fluctuations in both the frequency and value of liquidations, with pronounced spikes during periods of market stress. A sharp decline in cryptocurrency prices in March 2020, coinciding with the onset of COVID-19-related market disruptions, triggered approximately

Table 5

Liquidation Descriptives This table presents descriptive statistics on vault liquidations across various collateral types. *Liquidations* indicate the total number of liquidation events, while *Liquidations (Vault)* represent the number of unique user vault liquidations. *Debt Generated (Liquidated)* refers to the dollar value of debt that was generated and subsequently liquidated due to undercollateralization. *Discount* denotes the proportional discount applied to collateral during auctions to recover the outstanding debt. We report both an equally-weighted discount and a value-weighted discount, the latter accounting for the value auctioned within each collateral category. *Collateral Sold* reflects the proportion of collateral liquidated to cover debt and associated fees, with both equal-weighted and value-weighted estimates provided.

Collateral	Liquidations	Liquidations (Vault)	Debt Generated	Debt Liquidated	Discount (EW)	Discount (VW)	Collateral Sold (EW)	Collateral Sold (VW)
AAVE-A	38	36	17.041	2.480	0.023	0.001	0.544	0.575
BAL-A	13	11	5.776	0.296	0.030	0.020	0.493	0.427
BAT-A	239	224	19.512	3.083	0.122	0.022	0.248	0.280
COMP-A	16	16	14.259	0.914	0.034	0.010	0.609	0.538
ETH-A	3017	2793	13,371.217	201.377	0.022	0.087	0.777	0.418
ETH-B	90	88	1959.921	64.758	0.033	0.014	0.851	0.529
ETH-C	231	220	2582.718	31.539	0.013	0.015	0.667	0.606
GUSD-A	5	4	10.366	0.343	0.055	0.012	0.792	0.558
KNC-A	35	32	2.170	0.088	0.007	−0.004	0.249	0.528
LINK-A	141	115	422.913	13.540	0.007	0.000	0.657	0.640
LRC-A	30	25	12.021	1.403	0.023	0.081	0.542	0.304
MANA-A	13	12	23.544	0.157	0.042	0.039	0.690	0.712
MATIC-A	7	5	27.970	0.330	0.016	0.011	0.553	0.447
PAXUSD-A	9	9	129.927	6.172	0.040	0.004	0.917	1.000
RENBTC-A	29	29	25.346	1.793	0.006	0.008	0.632	0.652
RETH-A	1	1	46.225	0.166	0.005	0.005	0.679	0.679
TUSD-A	33	31	71.204	26.253	0.001	0.001	0.940	0.907
UNI-A	70	64	94.684	7.353	0.031	0.013	0.668	0.559
USDC-A	248	248	1697.245	223.683	0.005	0.002	0.812	0.782
USDT-A	9	9	2.678	0.001	0.011	0.011	0.578	0.589
WBTC-A	300	276	3862.139	77.402	0.026	0.081	0.760	0.316
WBTC-B	12	12	98.190	1.303	0.016	0.013	0.897	0.891
WBTC-C	17	16	411.860	1.514	0.007	−0.008	0.636	0.562
WSTETH-A	47	47	1078.894	5.384	0.005	−0.024	0.722	0.699
WSTETH-B	1	1	930.304	0.176	0.094	0.094	0.682	0.682
YFI-A	44	39	392.652	2.540	0.026	0.011	0.665	0.661
ZRX-A	35	31	9.882	5.335	0.102	0.152	0.450	0.240
Sum/Mean	4730	4394	27,360.072	679.381	0.023	0.076	0.732	0.584

1500 liquidations in that month alone.²³ While subsequent market corrections have generally resulted in fewer liquidations, the monetary value of these events has increased significantly, reflecting the broader growth of the DeFi ecosystem. Notably, liquidation values surpassed \$200 million in late 2022 following the collapse of FTX. Panel B shows the ratio of liquidated debt to the total debt generated across all vaults over the same period. This ratio peaked at around 8 % in early 2020, before declining to about 0.5 % by February 2021. Since then, it has gradually increased, stabilizing between 2.5 and 3 % by late 2023. This trend suggests that while the absolute value of liquidations has increased, users have become more resilient relative to the scale of overall lending activity.

5.3. Leverage and skill level

Table 6 summarizes the daily vault performance by leverage quintile and user skill level. Panel A reports vault return performance, while Panel B displays hypothetical returns performance in the absence of liquidation costs. Our analysis shows that average vault performance is positive and stable at around 0.3 % per day under low leverage. However, performance deteriorates as leverage increases, declining to 0.085 % per day in the highest quintile. This decline is statistically significant at the 1 % level.

Focusing on the level of user skill, it is evident that skilled users are more adept at using low to medium leverage. They consistently outperform unskilled users across all quintiles, with statistically significant differences observed in the lower leverage categories. Panel B further shows that, in the absence of liquidation costs, skilled users maintain a similar performance advantage at lower leverage levels, though the gap narrows in the highest quintile. This suggests that liquidations are largely unpredictable, and skilled users derive their advantage primarily during normal market conditions. The stark contrast between hypothetical and actual returns in the highest quintile (0.273 % vs 0.085 %) highlights the substantial impact of liquidation costs on high-leverage positions. Panel C shows the proportion of vault liquidations across leverage quintiles, disaggregated by user skill level. Forced liquidations are nearly nonexistent in the lower quintiles but rise sharply in the highest quintile reaching 0.263 % for skilled users and 0.279 % for unskilled users. This represents a tenfold increase from the fourth quintile. None of the quintiles exhibit a statistically significant difference in the proportion of forced liquidations between skilled and unskilled users. For comparison, Subrahmanyam et al. (2024) report forced liquidations as high as 1.74 % for unskilled users in futures markets, with the difference in liquidation probability between skilled and unskilled participants being statistically significant.

To examine the vault performance within a multivariate framework and explore the relation between user skill, leverage, and returns in greater depth, we estimate the following linear regression model:

$$\begin{aligned} Performance_{it} = & \beta_0 + \beta_1 DebtActions_{i,t-1} + \beta_2 VaultAge_{i,t-1} + \sum_{Q=2}^5 \beta_{3Q} D_{Q,i,t-1} \\ & + \sum_{Q=2}^5 \beta_{4Q} (D_{Q,i,t-1} \times Unskilled_i) + \alpha_i + \gamma_t + \epsilon_{it} \end{aligned} \quad (6)$$

where *Performance* is the daily vault return of vault *i* on day *t*. *Debt Actions* captures the number of debt-related transactions executed on that day. *Vault Age* denotes the age of the vault in years. $D_{Q,i,t-1}$ is a dummy equal to 1 if vault *i* at *t* – 1 is in LTV quintile *Q* (*Q* = 2, ..., 5; *Q* = 1, omitted as base), and *Unskilled_i* is a binary indicator for unskilled users. The interaction term *LTV* × *Unskilled* captures the differential effect of leverage across user skill levels. The regression includes vault and day fixed effects, with standard errors clustered at both levels.

Table ?? presents regression results analyzing the relationship between leverage, user skill, and vault performance. Column 2 displays overall performance across leverage quintiles, while columns 3 and 4 delineate results for skilled and unskilled users, respectively. The findings demonstrate that lower leverage levels do not significantly impair performance, whereas higher leverage levels prove detrimental. Specifically, vaults in the second-highest quintile (LTV_4) experience a performance reduction of 7.4 basis points (bps), which deepens to 33.3 bps in the highest quintile (LTV_5). The differential impact of leverage across user skill levels is particularly noteworthy. In the highest leverage quintile, unskilled users suffer a performance decline of 40.1 bps, compared to a smaller, though still significant, 27 bps reduction for skilled users. Interestingly, skilled users exhibit a positive relationship between the frequency of debt actions and performance (8.8 bps), suggesting that skill combined with active portfolio monitoring may partially mitigate the adverse effects of leverage. These results reinforce the univariate findings that leverage is detrimental to portfolio performance and that skilled users manage leverage more effectively than their unskilled counterparts, but both groups ultimately experience negative returns at higher leverage levels.

A particularly notable finding is the large baseline performance gap between skilled and unskilled users, as reflected in the constant terms (0.681 for skilled vs 0.393 for unskilled users). This 28.8 bp difference indicates that skilled users maintain significant performance advantages that is independent of leverage choices or portfolio management activities. This performance gap persists even in the hypothetical scenario without liquidation costs (0.600 vs 0.210), suggesting that the advantage of skilled users extends beyond

²³ It should be noted that during this period, Maker operated under a different liquidation framework than the one currently in use. During the ‘Black Thursday’ event in March 2020, the system permitted partial liquidations, allowing only a portion of a vault’s collateral to be sold to cover outstanding debt. This often resulted in multiple liquidation events for a single vault within the same day. Another key difference was the use of an English auction format for collateral actions, a mechanism that was later replaced by a Dutch auction mechanism with the introduction of Liquidations 2.0 in May 2021.

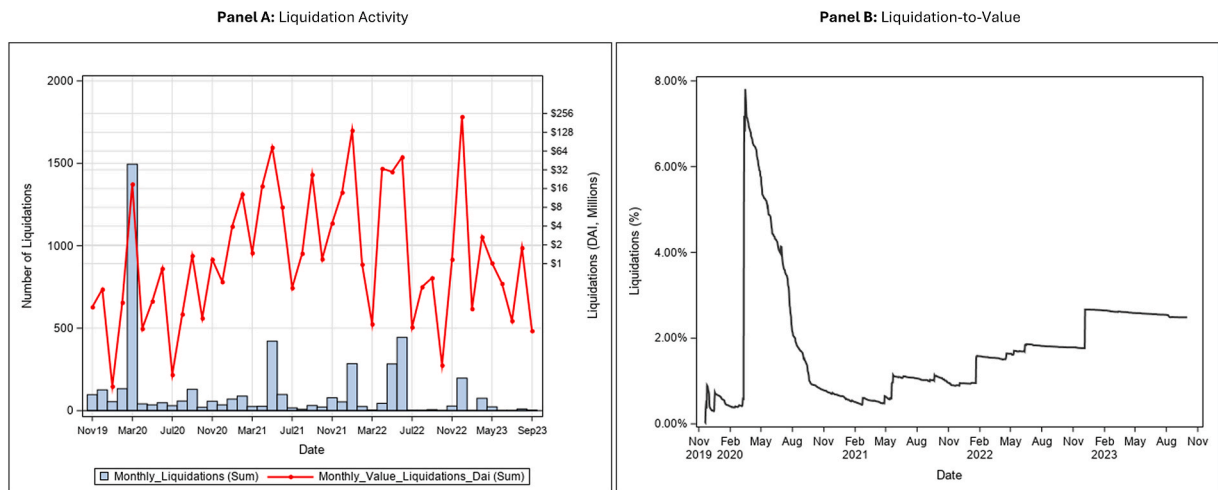


Fig. 4. Liquidations This figure presents liquidation activity over time and its value relative to outstanding debt. Panel A illustrates the monthly time series of liquidation activity in DeFi vaults from November 2019 to September 2023. The left axis indicates the number of liquidations, while the right axis shows their value in millions of Dai. The panel captures fluctuations, highlighting spikes during periods of market stress. Panel B illustrates the share of liquidated debt relative to the total debt generated across all vaults during the same period.

merely avoiding forced liquidations and reflects broader decision-making superiority. These findings are consistent with prior research in other financial markets. For example, [Subrahmanyam et al. \(2024\)](#) demonstrate that leverage negatively affects performance for all users in intraday futures trading, while [Davydov & Peltomäki \(2023\)](#) show that more attentive users achieve better outcomes from leverage in traditional equity margin trading. Further insights are provided in Column 5 of [Table 7](#), where interaction terms between

Table 6

Vault Performance by Leverage and Skill This table summarizes vault returns, hypothetical returns, and liquidation rates across leverage quintiles. Panel A presents the average daily returns for the sample, as well as for skilled and unskilled user groups. Panel B shows the hypothetical returns users would have been earned within each leverage quintile if no liquidation had occurred. Panel C reports the proportion of liquidations per leverage quintile. The last two columns compare forced liquidation rates between skilled and unskilled users, including p-values for testing the null of no difference. The last two rows report differences between the extreme quintiles and their corresponding p-values. Leverage quintiles are calculated on day t-1, all other variables are measured on day t. All returns are expressed as percentages.

LTV Group	Overall		Skilled		Unskilled		Unskilled - Skilled	
	LTV	Liquidate	LTV	Liquidate	LTV	Liquidate	Diff	p-value
Panel A: Actual Vault Returns								
Low	10.982	0.306	10.757	0.340	11.284	0.261	-0.079	<0.0001
2	18.873	0.333	18.966	0.369	18.759	0.289	-0.080	<0.0001
3	23.242	0.316	23.397	0.332	23.050	0.295	-0.037	0.031
4	27.137	0.282	27.343	0.287	26.897	0.278	-0.009	0.603
High	32.373	0.085	32.477	0.100	32.254	0.068	-0.032	0.097
High - Low	21.391	-0.221	21.720	-0.240	20.970	-0.193		
p-value	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001		
Panel B: Hypothetical Vault Returns								
Low	10.982	0.307	10.757	0.340	11.284	0.262	-0.078	<0.0001
2	18.873	0.334	18.966	0.370	18.759	0.291	-0.079	<0.0001
3	23.242	0.320	23.397	0.335	23.050	0.301	-0.034	0.046
4	27.137	0.298	27.343	0.299	26.897	0.296	-0.003	0.865
High	32.373	0.273	32.477	0.280	32.254	0.265	-0.015	0.364
High - Low	21.391	-0.034	21.720	-0.060	20.970	0.003		
p-value	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001		
Panel C: Liquidations								
Low	10.982	0.005	10.757	0.005	11.284	0.005	0.000	0.977
2	18.873	0.003	18.966	0.002	18.759	0.004	0.002	0.510
3	23.242	0.007	23.397	0.005	23.050	0.010	0.005	0.248
4	27.137	0.026	27.343	0.021	26.897	0.031	0.011	0.158
High	32.373	0.270	32.477	0.263	32.254	0.279	0.016	0.493
High - Low	21.391	0.265	21.720	0.258	20.970	0.274		
p-value	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001		

Table 7

Performance Regression Results This table presents linear regression results estimating vault returns. Columns (2–5) report actual performance, while columns (6–9) report hypothetical returns excluding liquidation costs. Key independent variables include # *Debt Actions* (frequency of debt-related transactions), *Vault Age* (in years), *LTV* quintiles (LTV ranks from low (2) to high (5)), and interaction terms (*LTV* × *Unskilled*) capturing the effect of unskilled management across LTV quintiles. *Unskilled* is an indicator variable identifying underperforming vaults based on pre-sample metrics. All models include vault and day fixed effects, with standard errors clustered at both levels. Coefficients are reported in percentage points. Statistical significance at the 10 %, 5 %, and 1 % levels is denoted by *, **, and ***, respectively.

	Vault Performance Returns				Vault Hypothetical Returns			
	Overall	Skilled	Unskilled	Skilled + Unskilled	Overall	Skilled	Unskilled	Skilled + Unskilled
# Debt Actions	0.111** (0.050)	0.088*** (0.031)	0.120 (0.083)	0.102** (0.050)	0.018 (0.029)	0.044** (0.018)	0.025 (0.037)	0.034 (0.025)
Vault Age	−0.224*** (0.085)	−0.403** (0.195)	−0.068 (0.114)	−0.239** (0.118)	−0.039 (0.062)	−0.300** (0.151)	0.195* (0.114)	−0.074 (0.096)
LTV_2	0.019 (0.014)	0.016 (0.012)	0.010 (0.022)	0.015 (0.013)	0.012 (0.010)	0.018 (0.011)	−0.002 (0.015)	0.018 (0.012)
LTV_3	−0.003 (0.017)	−0.002 (0.017)	−0.023 (0.026)	−0.003 (0.017)	−0.001 (0.015)	0.002 (0.016)	−0.021 (0.021)	0.002 (0.017)
LTV_4	−0.074*** (0.021)	−0.055*** (0.020)	−0.108*** (0.030)	−0.057*** (0.021)	−0.041** (0.019)	−0.035* (0.021)	−0.061** (0.027)	−0.035 (0.021)
LTV_5	−0.333*** (0.055)	−0.270*** (0.042)	−0.401*** (0.073)	−0.272*** (0.042)	−0.099*** (0.029)	−0.087*** (0.029)	−0.143*** (0.041)	−0.088*** (0.029)
LTV_2 x Unskilled				−0.004 (0.018)				−0.019 (0.013)
LTV_3 x Unskilled				−0.017 (0.019)				−0.022 (0.014)
LTV_4 x Unskilled				−0.048** (0.019)				−0.026* (0.016)
LTV_5 x Unskilled				−0.125*** (0.046)				−0.056** (0.023)
Constant	0.491*** (0.072)	0.681*** (0.166)	0.393*** (0.089)	0.531*** (0.097)	0.354*** (0.054)	0.600*** (0.129)	0.210** (0.085)	0.406*** (0.081)
Observations	3,530,927	1,157,204	921,543	2,078,747	3,530,927	1,157,204	921,543	2,078,747
R-squared	0.823	0.84	0.804	0.824	0.881	0.890	0.861	0.877
Vault Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Day Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

leverage quintiles and user skill reveal that unskilled users experience disproportionately larger performance declines at higher LTV levels, with an additional performance reduction of 12.8 bps. Conversely, lower LTV levels (LTV_2 and LTV_3) have minimal impact on performance for both skilled and unskilled users.

The interaction terms between leverage quintiles and user skill reveal a progressive pattern that deepens our understanding of how leverage interacts with investor ability. At lower leverage levels (LTV_2 and LTV_3), performance differences between skilled and unskilled users are modest and statistically insignificant. However, as leverage increases, the performance gap widens considerably. In the highest leverage quintile, unskilled users underperform skilled users by 12.5 percentage points, a difference that is significant at the 1 % level. This pattern demonstrates that skill deficits combined with high leverage create a compounding negative effect, with unskilled users experiencing disproportionately larger performance deterioration as leverage increases. These findings suggest that unskilled users have a lower optimal leverage threshold and would benefit from more conservative leverage strategies.

Columns 6–9 of Table 7 present a counterfactual analysis of hypothetical portfolio performance excluding liquidation costs. This approach assesses whether the poor performance of highly leveraged portfolios primarily stems from forced liquidation costs and whether skilled users are more effective at avoiding them. While negative performance persists in the higher quintiles, the magnitude of losses is substantially reduced in the absence of liquidation costs. In the highest quintile, skilled users experience a performance decline of 8.7 basis points, while unskilled users see a drop of 14.3 bps, significantly lower than the actual declines of 27 and 40.1 bps, respectively. These results indicate that forced liquidations account for more than half of the observed underperformance in highly leveraged portfolios. However, even absent these costs, high leverage continues to amplify losses, indicating that retail users generally struggle to enhance returns through market timing under high leverage, supporting the prior evidence from Heimer & Simsek (2019) and Heimer & Imas (2022). Importantly, the results also show that active debt management, such as adding collateral, repaying debt, or adjusting leverage, improves vault performance even when liquidation costs are excluded. This suggests that proactive portfolio adjustments play a critical role in mitigating risk, particularly during periods of heightened market volatility.

To investigate the determinants of forced liquidations, we employ a linear probability model, following the approach of Subrahmanyam et al. (2024), due to its straightforward interpretation of the marginal effects of explanatory variables. The model is specified as follows:

$$FL_{it} = \beta_0 + \sum_{Q=2}^5 \beta_{1Q} D_{Q,i,t-1} + \sum_{Q=2}^5 \beta_{2Q} (D_{Q,i,t-1} \times Unskilled_i) + \sum_{k=1}^K \beta_{3k} X_{k,i,t-1} + \alpha_i + \gamma_t + \epsilon_{it} \quad (7)$$

Table 8

Linear Probability Model Results This table presents regression coefficients from a linear probability model analyzing forced-liquidation events. The dependent variable equals one if a vault experiences forced liquidation on a given date and zero otherwise. Key independent variables include *# Debt Actions* (count of debt-related transactions), *Vault Age* and its square, *Loan-to-Value (LTV)* quintiles (2–5), and the interaction term $LTV \times Unskilled$. All specifications include vault and day fixed effects. Coefficients are marginal effects in percentage points; robust standard errors double-clustered by vault and date appear in parentheses. Significance: * 10 %, ** 5 %, *** 1 %.

	Overall	Skilled	Unskilled	Skilled + Unskilled
# Debt Actions	-0.039 (0.055)	-0.034 (0.049)	-0.020 (0.087)	-0.027 (0.057)
Vault Age	0.492*** (0.142)	0.484** (0.199)	0.792*** (0.231)	0.578*** (0.191)
Vault Age ²	-0.050*** (0.012)	-0.043*** (0.011)	-0.048*** (0.016)	-0.047*** (0.013)
LTV_2	-0.006 (0.006)	-0.002 (0.006)	-0.003 (0.009)	-0.001 (0.006)
LTV_3	0.001 (0.006)	0.000 (0.007)	0.014 (0.011)	0.001 (0.007)
LTV_4	0.040*** (0.012)	0.030*** (0.009)	0.062*** (0.017)	0.031*** (0.010)
LTV_5	0.352*** (0.073)	0.306*** (0.067)	0.382*** (0.083)	0.306*** (0.067)
LTV_2 × Unskilled				-0.003 (0.008)
LTV_3 × Unskilled				0.012 (0.011)
LTV_4 × Unskilled				0.029** (0.014)
LTV_5 × Unskilled				0.073* (0.040)
Constant	-0.306*** (0.100)	-0.317** (0.145)	-0.487*** (0.150)	-0.368*** (0.132)
Observations	3,530,927	1,157,204	921,543	2,078,747
R-squared	0.049	0.042	0.057	0.048
Vault Fixed Effects	Yes	Yes	Yes	Yes
Day Fixed Effects	Yes	Yes	Yes	Yes

where FL_{it} is a binary forced liquidation indicator, $D_{Q,i,t-1}$ are LTV quintiles (previously defined), $X_{k,i,t-1}$ are lagged control variables, including *Debt Actions*, representing the number of debt-related transactions, *Vault Age* along with its squared term, capturing the operational lifespan of a vault and potential nonlinear effects, and all other notation as previously defined.

The regression results are reported in Table 8. As expected, LTV is positively related to forced liquidations, while the number of debt actions exhibits a negative relationship, though not statistically significant. The coefficients for vault age and its squared term are statistically significant across all specifications, indicating a nonlinear relationship between operational lifespan and liquidation risk. Notably, vault age consistently exhibits a positive and highly statistically significant coefficient, suggesting that older vaults face greater liquidation risks. This finding implies that accumulated market experience does not necessarily shield users from liquidation events. The effect of vault age also varies by user type. Unskilled users show a coefficient of 0.792, compared to 0.484 for skilled users, indicating that unskilled users become increasingly vulnerable to liquidation as their vaults age. In contrast, the squared term of vault age is negative and highly significant, suggesting diminishing marginal effects over time. Higher LTV ratios, particularly LTV_4 and LTV_5, are strongly and positively associated with forced liquidations. Being in the highest leverage quintile increases liquidation probability by 0.352 % in the full specification. These results underscore the critical role of excessive leverage in driving liquidation events, consistent with theoretical expectations regarding the risks inherent in collateralized debt positions.

A clear pattern emerges when examining the interaction between leverage and user skill. The interaction effects between LTV quintiles and the unskilled user indicator exhibit a progressive increase in magnitude, with $LTV_4 \times Unskilled$ (0.029) and $LTV_5 \times Unskilled$ (0.073) reaching statistical significance at the 5 % and 10 % levels, respectively. Although the statistical support is modest, the pattern aligns with the performance regression results in Table 7, where skill differences became increasingly pronounced at higher leverage levels. These findings suggest that unskilled users not only experience greater performance deterioration under high leverage but also face systematically higher probabilities of forced liquidation. This compounding effect helps explain the consistently poorer outcomes observed among unskilled users in high-leverage environments.

6. Conclusion

This study explores a novel DeFi mechanism that enables the use of crypto assets as collateral for autonomous, non-contingent lending arrangements. We provide a comprehensive overview and examination of one of the largest trustless lending systems within the crypto-asset market, focusing on loan demand, leverage and performance dynamics, and the role of liquidations. We further examine whether leverage and performance outcomes vary by user skill, using an identification period to classify trader skill. Our

analysis reveals several key insights. First, we show that protocol parameters (including stability fees and liquidation ratios) have a significant impact on loan demand at an aggregated collateral level. Notably, we observe a positive relationship between stability fees and loan demand. This finding suggests the presence of yield-seeking behavior and a willingness to incur higher costs for liquidity may drive borrower decisions in the DeFi market. Importantly, our results underscore the material impact of protocol-specific and blockchain-inherent frictions, such as the negative correlation observed between gas fees and borrowing activity, and the sensitivity of loan demand to the stability of the protocol's native stablecoin, Dai.

The second part of this paper focuses on individual vault-level analysis, specifically examining how leverage influences trading performance across the cross-section of users. Our findings indicate that, overall, leverage is associated with diminished trading performance and elevated liquidation risk. However, the extent of this impact varies significantly by user skill level. Skilled users are generally able to manage moderate leverage effectively, whereas excessive leverage proves detrimental across the board—particularly for less sophisticated users. The liquidation mechanism, while essential for maintaining protocol stability, acts as a powerful amplifier of negative outcomes. We find moderate evidence that forced liquidations disproportionately affect unskilled users, underscoring a critical tension in decentralized lending systems: the need to balance protocol security with the protection of vulnerable participants. These findings carry important implications for various stakeholders. For users, they highlight the necessity of education and robust risk management when engaging with DeFi protocols. For protocol designers, the results emphasize the importance of implementing advanced risk controls and user interfaces that help safeguard less experienced users from the dangers of excessive leverage.

Appendix A. Maker Collateral Tokens

Table A1

Tokens used as collateral This table lists every token that has been onboarded to the Maker system since inception. Onboarding begins with a MakerDAO community application, followed by a Community Greenlight Poll, Core-Unit risk assessments, a Governance Poll, and an executive vote. Off-boarding reverses the process, gradually reducing exposure and debt ceilings while safeguarding the stability of DAI.

Token	Name	Category	Description
AAVE	Aave	Utility	Governance token for Aave, a decentralized liquidity protocol (added Dec 2020).
BAL	Balancer	Utility	Governance token for Balancer liquidity protocol (added Oct 2020).
BAT	Basic Attention Token	Utility	Reward token for the Brave browser (added Nov 2019).
COMP	Compound	Utility	Governance token for Compound Finance (added Sep 2020).
ETH	Ethereum	Utility	Native coin of Ethereum; most common collateral for DAI.
GNO	Gnosis	Utility	Governance/staking token for Gnosis (added Sep 2022).
KNC	Kyber Network Crystal	Utility	Fee token for Kyber Network DEX (added Jun 2020).
LINK	Chainlink	Utility	Payment token for Chainlink oracle network (added Sep 2020).
LRC	Loopring	Utility	Native token for Loopring protocol (added Sep 2020).
MANA	Decentraland	Utility	Native currency of the Decentraland metaverse (added Jul 2020).
MATIC	Polygon	Utility	Settlement token for Polygon network (added Aug 2021).
RENBTC	renBTC	Utility	ERC-20 token backed 1:1 by BTC via RenVM (added Dec 2020).
RETH	Rocket Pool ETH	Utility	Liquid-staking derivative for Rocket Pool staked ETH (added Nov 2022).
UNI	Uniswap	Utility	Governance token for Uniswap DEX (added Dec 2020).
WBTC	Wrapped Bitcoin	Utility	BTC-backed ERC-20 issued by BitGo (added May 2020).
WSTETH	Wrapped Staked Ether	Utility	Non-rebasing wrapper of stETH from Lido (added May 2022).
YFI	Yearn Finance	Utility	Governance token for Yearn Finance aggregator (added Oct 2020).
ZRX	Ox	Utility	Governance token for the Ox protocol (added Jun 2020).
GUSD	Gemini Dollar	Stablecoin	Regulated USD-backed stablecoin issued by Gemini (added Nov 2020).
USDP	Pax Dollar	Stablecoin	Regulated USD-backed stablecoin issued by Paxos (added Sep 2020).
TUSD	TrueUSD	Stablecoin	USD-backed stablecoin issued by TrustToken (added Jun 2020).
USDC	USD Coin	Stablecoin	USD-backed stablecoin issued by Circle/Coinbase (added Mar 2020).

Appendix B. Stability Fees and Total Debt

Vaults with outstanding debt are charged a stability fee by the Maker protocol. All vaults incur a base rate (r_f), and an additional duty rate (r_p), or risk premium, based on the type of collateral locked. These rates are combined in a single rate accumulator. Since each vault (v) is tied to a specific collateral type, the cumulative rate $R(v, t)$ can be expressed as:

$$R(v, t) = R(v, 0) \times \prod_{i=t_0+1}^t (1 + F_i) \quad (1)$$

where F_i is the per-second stability fee at time i , accumulated in 1-s intervals from t_0 . For small intervals, F_i approximates the continuous rate $r_f + r_p$ as follows:

$$F_i = (r_f + r_p)\Delta t, \quad \text{with} \quad \Delta t = 1 \text{ second} \quad (2)$$

Thus, for small Δt , we have:

$$1 + F_i \approx e^{(r_f + r_p)\Delta t} \quad (3)$$

Accordingly, the product per-second rates approximates the continuous exponential function:

$$\prod_{i=t_0+1}^t (1 + F_i) \approx \exp\left(\sum_{i=t_0+1}^t (r_f + r_p)\Delta t\right) = e^{(r_f + r_p)(t-t_0)} \quad (4)$$

Therefore, the cumulative rate can also be expressed as:

$$R(v, t) = R(v, 0) \times e^{(r_f + r_p)(t-t_0)} \quad (5)$$

Let a vault be created at time t_0 with debt $D(v, 0)$ drawn immediately. The normalized debt $A(v, 0)$ is:

$$A(v, 0) = \frac{D(v, 0)}{R(v, 0)} \quad (6)$$

Substituting the expressions for $A(v, 0)$ and $R(v, t)$, we get:

$$D(v, t) = A(v, 0) \times R(v, t) = \frac{D(v, 0)}{R(v, 0)} \times R(v, 0) \times e^{(r_f + r_p)(t-t_0)} \quad (7)$$

Simplifying, we obtain an expression for the total debt:

$$D(v, t) = D(v, 0) \times e^{(r_f + r_p)(t-t_0)} \quad (8)$$

Data availability

Data will be made available on request.

References

- Acharya, V. V., & Viswanathan, S. (2011). Leverage, moral hazard, and liquidity. *The Journal of Finance*, 66(1), 99–138.
- Adrian, T., & Shin, H. S. (2010). Liquidity and leverage. *Journal of Financial Intermediation*, 19(3), 418–437.
- Adrian, T., & Shin, H. S. (2014). Procyclical leverage and value-at-risk. *Review of Financial Studies*, 27(2), 373–403.
- Aquilina, M., Foley, S., Gambacorta, L., & Krekel, W. (2024a). Decentralised dealers? Examining liquidity provision in decentralised exchanges. *Bank for international settlements BIS working papers* (p. 1227).
- Aquilina, M., Frost, J., & Schrimpf, A. (2024). Decentralized finance (defi): A functional approach. *Journal of Financial Regulation*, 10(1), 1–27.
- Aramonte, S., Doerr, S., Huang, W., & Schrimpf, A. (2022). *Defi lending: Intermediation without information?* (Vol. 57). BIS Bulletins.
- Aspris, A., Dyhrberg, A. H., Putnig, T. J., & Foley, S. (2022). *Digital assets and markets: A transaction-cost analysis of market architectures*. SWIFT Institute Working Paper.
- Barber, B. M., Huang, X., Ko, K., & Odean, T. (2020). Leveraging overconfidence. Available at: SSRN: <https://ssrn.com/abstract=3445660>.
- Barberis, N., Shleifer, A., & Vishny, R. (1998). A model of investor sentiment. *Journal of Financial Economics*, 49(3), 307–343.
- Barbon, A., & Rinaldo, A. (2021). *On the quality of cryptocurrency markets: Centralized versus decentralized exchanges*. Papers 2112.07386, arXiv.org.
- Benmelech, E., & Bergman, N. K. (2009). Collateral pricing. *Journal of Financial Economics*, 91(3), 339–360.
- Bernanke, B. S., & Blinder, A. S. (1988). Credit, money, and aggregate demand. *The American Economic Review*, 78(2), 435–439.
- Bernanke, B. S., & Gertler, M. (1995). Inside the Black box: The credit channel of monetary policy transmission. *The Journal of Economic Perspectives*, 9(4), 27–48.
- Bian, J., Da, Z., He, Z., Lou, D., Shue, K., & Zhou, H. (2024). The drivers and implications of retail margin trading. *The Journal of Finance* (in press).
- BIS. (2018). Sound practices: Implications of fintech developments for banks and bank supervisors. <https://www.bis.org/bcbs/publ/d431.pdf>.
- Campello, M., Jin, P., Rabetti, D., & Saleh, F. (2023). *The market for crypto zombies: Under-collateralization in defi lending*. Working Paper.
- Capponi, A., & Jia, R. (2021). *The adoption of blockchain-based decentralized exchanges*. arXiv preprint arXiv:2103.08842.
- Cheah, E.-T., & Fry, J. (2015). Speculative bubbles in bitcoin markets? An empirical investigation into the fundamental value of bitcoin. *Economics Letters*, 130, 32–36.
- Chen, J. (2019). The impact of controlling shareholder equity pledge on rd investment. *American Journal of Industrial and Business Management*, 9, 2129–2143.
- Chen, Z., Li, P., Wang, Z., & Zhang, B. (2024). Leveraged trading and stock returns: Evidence from international stock markets. *Journal of Financial Markets*, 69, Article 100907.
- Chiu, J., Ozdenoren, E., Yuan, K., & Zhang, S. (2024). On the fragility of defi lending. Available at: SSRN: <https://ssrn.com/abstract=4328481>.
- Cornelli, G., Gambacorta, L., Garratt, R., & Reghezza, A. (2024). *Why defi lending?* CEPR discussion papers 19358, C.E.P.R. discussion papers.
- Davydov, D., & Peltomäki, J. (2023). Investor attention and the use of leverage. *The Financial Review*, 2, 287–313.
- De Blasis, R., & Webb, A. (2022). Arbitrage, contract design, and market structure in bitcoin futures markets. *Journal of Futures Markets*, 42(3), 492–524.
- Frazzini, A., & Pedersen, L. H. (2022). Embedded leverage. *The Review of Asset Pricing Studies*, 12(1), 1–52.
- Gadzinski, G., Castello, A., & Mazzorana, F. (2023). Stablecoins: Does design affect stability? *Finance Research Letters*, 53, Article 103611.
- Geanakoplos, J. (2001). Liquidity, default and crashes: Endogenous contracts in general equilibrium. *Cowles foundation discussion papers 1316*. Cowles Foundation for Research in Economics, Yale University.
- Geanakoplos, J. (2010). The leverage cycle. *NBER Macroeconomics Annual*, 24(1), 1–66.
- Hackethal, A., Hanspal, T., Lammer, D. M., & Rink, K. (2022). The characteristics and portfolio behavior of bitcoin investors: Evidence from indirect cryptocurrency investments. *Review of Finance*, 26(4), 855–898.
- Han, J., Lee, J., & Li, T. (2025). A review of dao governance: Recent literature and emerging trends. *Journal of Corporate Finance*, 91, Article 102734.
- Harvey, C. R., Abou Zeid, T., Draaisma, T., Luk, M., Neville, H., Rzym, A., & Van Hemert, O. (2022). An investor's guide to crypto. *Journal of Portfolio Management*, 1 (49), 146–171.
- Heimbach, L., & Huang, W. (2024). *Defi leverage*. BIS Working Papers, No 1171.

- Heimer, R. Z., & Imas, A. (2022). Biased by choice: How financial constraints can reduce financial mistakes. *Review of Financial Studies*, 35(4), 1643–1681.
- Heimer, R., & Simsek, A. (2019). Should retail investors' leverage be limited? *Journal of Financial Economics*, 132(3), 1–21.
- IMF. (2022). The rapid growth of fintech: Vulnerabilities and challenges for financial stability. *International monetary fund global financial stability report*. April: (Chapter 3).
- Jones, M., Lutu, T. J., & Samuel, B. (2024). The interdependence of financial literacy and crypto literacy. *Economics Letters*, 239, Article 111737.
- Kogan, S., Makarov, I., Niessner, M., & Schoar, A. (2024). Are cryptos different? Evidence from retail trading. *Journal of Financial Economics*, 159, Article 103897.
- Kumar, A. (2009). Who gambles in the stock market? *The Journal of Finance*, 64, 1889–1933.
- Ladley, D., Liu, G., & Rockey, J. (2020). Losing money on the margin. *Journal of Economic Behavior & Organization*, 172, 107–136.
- Laeven, L., Levine, R., & Michalopoulos, S. (2015). Financial innovation and endogenous growth. *Journal of Financial Intermediation*, 24(1), 1–24.
- Lehar, A., & Parlour, C. A. (2022). Systemic fragility in decentralized markets. Available at: SSRN: <https://ssrn.com/abstract=4164833>.
- Lehar, A., & Parlour, C. A. (2024). Decentralized exchange: The uniswap automated market maker. *The Journal of Finance* (in press).
- Liberti, J. M., & Petersen, M. A. (2019). Information: Hard and soft. *The Review of Corporate Finance Studies*, 8(1), 1–41.
- Liu, Y., Tsyvinski, A., & Goldstein, I. (2021). Risks and returns of cryptocurrency. *Review of Financial Studies*, 34(6), 2689–2727.
- Liu, Y., Tsyvinski, A., & Wu, X. (2022). Common risk factors in cryptocurrency. *The Journal of Finance*, 77(2), 1133–1177.
- Lusardi, A., & Mitchell, O. S. (2023). The importance of financial literacy: Opening a new field. *The Journal of Economic Perspectives*, 37(4), 137–154.
- Makarov, I., & Schoar, A. (2022). Cryptocurrencies and decentralized finance (defi). *Brookings Papers on Economic Activity*, 1, 141–215.
- Momtaz, P. P. (2021). The pricing and performance of cryptocurrency. *The European Journal of Finance*, 27(4–5), 367–380.
- Mueller, P. (2024). Defi leveraged trading: Inequitable costs of decentralization. Available at: SSRN: <https://ssrn.com/abstract=4241356>.
- Odean, T. (1999). Do investors trade too much? *The American Economic Review*, 89(5), 1279–1298.
- Pang, C., & Wang, Y. (2020). Stock pledge, risk of losing control and corporate innovation. *Journal of Corporate Finance*, 60, Article 101534.
- Park, A., & Stinner, J. (2023). Phantom liquidity in decentralized lending. *SSRN Electronic Journal*.
- Saengchote, K. (2023). Decentralized lending and its users: Insights from compound. *Journal of International Financial Markets, Institutions and Money*, 87, Article 101807.
- Schär, F. (2021). Decentralized finance: On blockchain-and smart contract-based financial markets. *FRB of St. Louis Review*.
- Stiglitz, J. E., & Weiss, A. (1981). Credit rationing in markets with imperfect information. *The American Economic Review*, 71(3), 393–410.
- Stulz, R., & Johnson, H. (1985). An analysis of secured debt. *Journal of Financial Economics*, 14(4), 501–521.
- Subrahmanyam, A., Tang, K., Wang, J., & Yang, X. (2024). Leverage is a double-edged sword. *The Journal of Finance*, 74, 1579–1634.
- Tzur-Ilan, N. (2023). Adjusting to macroprudential policies: Loan-to-value limits and housing choice. *Review of Financial Studies*, 36(10), 3999–4044.
- Wang, Q. (2024). Does defi democratize access to financial services?. Available at: SSRN: <https://ssrn.com/abstract=4779624>.
- Yardeni, E., & Quintana, M. (2022). *Stock market indicators: Margin debt*. Research: Yardeni. <https://www.yardeni.com/pub/stmkteqmardebt.pdf>.
- Yermack, D. (2015). Bitcoin, innovation, financial instruments, and big data. *Handbook of digital currency*.
- Zhao, X., Wang, Y., & Liu, W. (2024). Someone like you: Lottery-like preference and the cross-section of expected returns in the cryptocurrency market. *Journal of International Financial Markets, Institutions and Money*, 91, Article 101957.